

CRAFT: Grounded Multi-Agent Coordination Under Partial Information

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Abstract

We introduce CRAFT, a multi-agent benchmark for evaluating pragmatic communication in large language models (LLMs) under strict partial information conditions. In this setting, multiple agents with complementary but incomplete views must coordinate through natural language to construct a single 3D structure whose final form is *unknown to any individual agent*. We formalize this as a multi-sender Bounded Pragmatic Speaker problem and provide an evaluation and diagnostic framework that decomposes failures into *spatial grounding*, *mind modeling* and *pragmatic communication* errors. Across 8 open-weight and 7 frontier models, we find that stronger reasoning ability does not reliably translate to better coordination, and improved individual communication does not guarantee successful collaboration. These results suggest that multi-agent coordination remains a fundamentally unsolved challenge for current LLMs, and CRAFT provides a novel framework to meaningfully advance evaluation in multi-agent collaboration. Source code and benchmark data: <https://github.com/csu-signal/CRAFT>

1 Introduction

Large language models (LLMs) have evolved from single-turn assistants into components of multi-agent systems where models must coordinate toward shared goals through open-ended natural language [Kazemitabaar et al., 2023, Zhang et al., 2024b, Cui et al., 2024, Davidson et al., 2025]. Yet coordination between LLM agents remains fragile even in simple settings [Lupu et al., 2021, Agashe et al., 2025, Singh et al., 2025, Nath and Krishnaswamy, 2025, Riedl, 2025, Davidson et al., 2025, Eisenstein et al., 2026]: models struggle with partner modeling and joint decision-making even when interacting with identical copies of themselves—and critically, even when trained with Reinforcement Learning from Human Feedback (RLHF) [Stiennon et al., 2020], which optimizes for human-preferred responses but does not explicitly train for the communicative demands of multi-agent coordination. This challenge compounds under *partial observability*, where agents hold differing private information and must decide what to communicate based on others’ knowledge states.

We argue that the missing capability is *pragmatic communication*—deciding what and how much to say and when to say it, based on other agents’ knowledge and needs [Grice, 1975, Frank and Goodman, 2012]. While frameworks such as Rational Speech Acts (RSA) [Goodman and Stuhlmüller, 2013] and the Bounded Pragmatic Speaker (BPS) model [Nguyen, 2024] formalize this behavior, existing evaluations largely focus on single-agent reasoning or offline interpretation [Jian and Siddharth,

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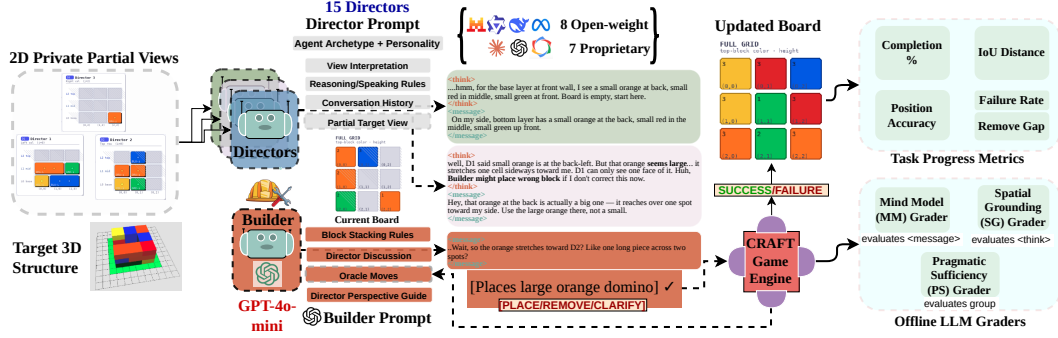


Figure 1: **CRAFT overview.** A structure generator creates a target 3D object and 3 private 2D views for Directors, enforcing information asymmetry. Each turn, Directors produce instructions from their partial views, and a Builder executes them in the CRAFT environment, which logs task progress and evaluates communication for **spatial grounding**, **mind modeling**, and **pragmatic sufficiency**.

2024, Zhu et al., 2026]. This creates a key gap: LLMs may exhibit strong internal reasoning (formal competence) but fail to use it effectively in interaction (functional competence) [Mahowald et al., 2024, Davidson et al., 2025].

Consider the Distributed Partial Information Puzzle (DPIP; Zhu et al. 2026): 3 **Directors**, each holding a distinct partial view of a 3D structure, must instruct a **Builder** to reconstruct it through natural language alone—a multi-party coordination structure our work directly adopts for LLMs. Humans succeed in DPIP through sustained pragmatic repair across multiple turns², producing precisely the complementary, non-redundant coordination that RSA and BPS theory predict—and that collective intelligence research identifies as necessary for group-level synergy [Humphreys, 1997].

Yet two critical questions remain unanswered. First, while [Nguyen, 2024] suggests RLHF-trained LLMs approximate bounded pragmatic speakers in dyadic settings, it is unknown whether this extends to multi-agent coordination where each agent simultaneously holds distinct private information and must model multiple partners’ knowledge states. Second, existing evaluations in pragmatic-collaborative tasks for LLMs rely on offline analysis [Ma et al., 2025a] or dyadic setups [Wu et al., 2024, Davidson et al., 2025]—leaving open whether LLM collectives exhibit the integrated pragmatic “synergy” humans achieve in DPIP, or instead fall into “collaboration gaps” [Davidson et al., 2025] that emerge when coordination under partial observability breaks down. CRAFT is designed to answer both: **does the pragmatic competence that BPS theory predicts and humans exhibit in collaborative tasks like DPIP transfer to LLM collectives operating under the same epistemic asymmetry, and if not, where and how does it break down in LLM-native multi-agent coordination?**

To answer this rigorously and at scale, we introduce three original contributions: a *procedural structure generator* with enforced information asymmetry; a *physically-grounded game engine* with ground-truth move validation and an oracle-assisted Builder interface to disentangle Builder capability from Director communication, enabling proper credit assignment in multi-agent systems [Huh and Mohapatra, 2024]; and a *3-judge evaluation framework* decomposing communication failures into spatial grounding of private reasoning, epistemic calibration of public messages to the group, and collective pragmatic sufficiency—each judge targeting a distinct failure mode predicted by our *multi-sender BPS formalization* of directors as bounded pragmatic speakers [Frank and Goodman, 2012, Nguyen, 2024]. Our evaluation across **15** frontier and open-weight LLMs reveals that scale *alone* does not confer reliable coordination advantages, that individual communication quality does not always lead to better collaboration, and that LLM collectives exhibit previously unknown behavioral patterns such as “correction spirals” under information asymmetry. These findings expose a fundamental gap between individual communicative competence in LLM agents and collective task success. Fig. 1 provides a high-level overview of CRAFT.

²Table 4 in Appendix C shows a snippet from a DPIP “game” where humans take ~30s and 23 turns of dialogic exchanges involving pragmatic repair to resolve a *single* action.

2 Related Works

Spatial reasoning and multi-agent coordination under partial observability Recent work has advanced spatial reasoning for LLMs across viewpoint consistency, 3D reasoning, and object localization [Kamath et al., 2023, Ma et al., 2025b, Du et al., 2024, Li et al., 2025, Liu et al., 2025, Zhang et al., 2025, Xu et al., 2026, Yeh et al., 2026], with extensions to dialogue and navigation [Bickmore and Cassell, 2005, Gao et al., 2022, Zhang et al., 2024a, Hou et al., 2025, Martorell, 2025, Zheng et al., 2025]. In parallel, LLM coordination has been studied in symmetric games, zero-shot partner matching, and agentic pipelines [Lupu et al., 2021, Jiang et al., 2024, Agashe et al., 2025, Grötschla et al., 2025, Maslej et al., 2025, Nath et al., 2025a,b, Nath and Krishnaswamy, 2025, Singh et al., 2025, Chen et al., 2026, Hayler et al., 2026]. However, spatial benchmarks are typically single-agent perception tasks, while coordination studies focus on symmetric or abstract settings with limited grounding in shared physical state [Liu et al., 2025, Mohammadi et al., 2025, Tran et al., 2025, Chen et al., 2026]. Real-world collaboration instead requires agents to operate under partial observability with complementary private evidence communicated over multiple turns [Chen et al., 2024, Liang et al., 2025], where coordination remains fragile even in simplified environments [Tian et al., 2020, Davidson et al., 2024, 2025, Grötschla et al., 2025, Wu et al., 2025] and demands informative, concise communication [Garrod and Pickering, 2004, Davidson et al., 2025]. Motivated by distributed partial information tasks [Zhu et al., 2026] and reference games [Nunberg, 1978, Andreas and Klein, 2016, Ma et al., 2025a], CRAFT addresses this gap by evaluating pragmatic communication in grounded multi-agent settings with asymmetric spatial information, measuring both task completion and communication quality [Amirkhani and Barshooi, 2022].

Pragmatic reasoning and bounded pragmatic speakers Pragmatic language use has been widely studied through Gricean accounts and Rational Speech Acts (RSA), where speakers select utterances by modeling a listener’s beliefs and informational needs [Grice, 1975, Frank and Goodman, 2012, Goodman and Stuhlmüller, 2013]. Prior work has explored pragmatics in controlled reference games, typically in two-agent settings [Fried et al., 2018, Khani et al., 2018, Nematzadeh et al., 2018, Louis et al., 2020, Hu et al., 2023, Ruis et al., 2023, Zhang et al., 2023, Estienne et al., 2025], while recent work extends this perspective to LLMs via the Bounded Pragmatic Speaker (BPS) framework [Nguyen, 2024]. Benchmarks such as DiPlomat show that LLMs struggle with context-sensitive implicature and situated interpretation [Li et al., 2023], and surveys highlight that existing evaluations emphasize pragmatic *understanding* over *production*, yet LLMs perform substantially better as listeners than as speakers [Ferreira, 2005, Meyer et al., 2016, Sieker and Zarriß, 2026, Krause and Vossen, 2024, Park et al., 2024]—a asymmetry that matters precisely because models may encode rich representations yet fail to deploy them in interaction [Mahowald et al., 2024, Wu et al., 2024, Davidson et al., 2025]. CRAFT targets this gap by separating the pragmatic “speaker” role from the “listener” role, isolating production as the operative challenge in a multi-agent partially observable setting where joint action depends on it.

3 CRAFT as a Multi-Sender Bounded Pragmatic Speaker Framework

The Bounded Pragmatic Speaker (BPS) framework [Nguyen, 2024] provides a unified account of pragmatic language production in LLMs, in which a *pragmatic speaker* selects utterances by jointly reasoning about a base generative model and a Theory-of-Mind (ToM) listener that evaluates how well an utterance communicates an intended meaning.

Definition 1 (Bounded Pragmatic Speaker). *Let $S_{base}(u \mid z^*, c)$ be a base speaker distribution over utterances $u \in \mathcal{U}$, conditioned on an intention $z^* \in \mathcal{Z}$ and context c . Let $L_{ToM}(z^* \mid u, c)$ be a Theory-of-Mind (ToM) listener that scores how faithfully u communicates z^* . A Bounded Pragmatic Speaker (BPS) selects utterances as:*

$$S_{bps}(u \mid z^*, c) \propto S_{base}(u \mid z^*, c) \cdot L_{ToM}(z^* \mid u, c). \quad (1)$$

Nguyen [2024] show that any language model S_θ can be viewed as a BPS by setting both modules to S_θ , and that RLHF fine-tuning [Stiennon et al., 2020, Tunstall et al., 2023] is equivalent to variational inference on a BPS whose ToM listener is a learned reward function R_ϕ . We extend this framework to the *multi-agent, grounded, partial-information* setting of CRAFT.

3.1 Directors as Bounded Pragmatic Speakers

Director D_i at turn t has an *intention* $z_{i,t}^* = \Delta_i(s_t, \mathcal{T})$ —the grounded gap between D_i ’s target view $\mathcal{T}$ and the current world state s_t (the observable board configuration at turn t)—and a *context*

$c_{i,t} = (o_{i,t}, h_t, \{u_{j,t}\}_{j \neq i})$ comprising their private observation, the conversation history, and the current-turn utterances of the other Directors. Under the BPS framework, D_i 's policy is:

$$\pi_{D_i}(u_{i,t} \mid z_{i,t}^*, c_{i,t}) \propto S_{\text{base}_i}(u_{i,t} \mid z_{i,t}^*, c_{i,t}) \cdot L_{\text{ToM}_i}(z_{i,t}^* \mid u_{i,t}, c_{i,t}), \quad (2)$$

where L_{ToM_i} models how the Builder interprets $u_{i,t}$ given the already-communicated context. However, the goal of each Director D_i is not merely to produce an utterance consistent with its private view $o_{i,t}$, but to select $u_{i,t}$ that maximizes its marginal contribution to a *joint* listener aggregating all three Directors—resolving the Builder's remaining uncertainty about z^* given $\{u_{j,t}\}_{j \neq i} \subset c_{i,t}$, such that the collective output moves s_t progressively toward the ground truth target $\mathcal{T}$.

Definition 2 (Joint ToM Listener). *The joint ToM listener for the Builder is:*

$$L_{\text{ToM}_{\text{joint}}}(z^* \mid u_1, u_2, u_3, c) \propto \exp\left(\sum_{i=1}^3 \lambda_i R_i(u_i, s_t, \mathcal{T})\right), \quad (3)$$

where $R_i(u_i, s_t, \mathcal{T})$ is a reward signal measuring the downstream task progress attributable to Director D_i 's utterance $u_{i,t}$, and $\lambda_i \geq 0$ are weighting coefficients.

This leads to the main theoretical result of this section.

Theorem 1 (CRAFT as a Multi-Sender BPS). *Under Eq. 1 and Eq. 3, let $S_{\text{base}_i}(\cdot \mid z_i^*, c_i)$ be the base speaker of director D_i and let $z^* = (z_1^*, z_2^*, z_3^*)$ be the joint intention vector; where $c_i = (o_{i,t}, h_t, \{u_{j,t}\}_{j \neq i})$ is director D_i 's private context and $c = (h_t, s_t)$ is the shared public context available to all agents. The optimal joint Director policy is equivalent to a **multi-sender Bounded Pragmatic Speaker**. Formally, the optimal joint Director policy satisfies:*

$$\pi^*(u_1, u_2, u_3 \mid z^*, c) \propto \left(\prod_{i=1}^3 S_{\text{base}_i}(u_i \mid z_i^*, c_i)\right) \cdot L_{\text{ToM}_{\text{joint}}}(z^* \mid u_1, u_2, u_3, c), \quad (4)$$

which reduces to the standard single-sender BPS (Definition 1) under $N = 1$ (see Appendix A for proofs.).

Theorem 1 establishes that optimal Director behavior requires each D_i to satisfy two simultaneous conditions: (i) faithfully encoding its private observations into candidate utterances via S_{base_i} , and (ii) calibrating those utterances to the *joint* information need of the Builder via $L_{\text{ToM}_{\text{joint}}}$ —contributing exactly the information the Builder requires that has not already been provided by the other Directors.³ Importantly, this condition is irreducible to any individual Director's output quality: a **communication failure** occurs when $L_{\text{ToM}_{\text{joint}}}$ fails to place sufficient mass on z^* despite it being reachable⁴, regardless of how well any individual Director reasoned or communicated.

Failure Modes in Multi-Sender BPS Nguyen [2024] identifies three failure modes for a single-sender BPS: **F1** (*limited search*), **F2** (*flawed pragmatics*), and **F3** (*inefficient inference*). In CRAFT, F1 occurs when S_{base_i} fails to identify the correct missing block from D_i 's visible wall; F3 occurs when correct private reasoning fails to transfer into a sufficiently specified public message. Both remain agent-local. F2 becomes strictly harder in the multi-sender setting: each Director must calibrate against $L_{\text{ToM}_{\text{joint}}}$ rather than a single listener, accounting for what all other Directors have already communicated. As implied in Theorem 1, a Director may avoid F1–F3 individually yet still contribute to a group-level failure—the irreducible fourth failure mode motivating the diagnostic framework in Section 5.3.

4 Agent Architecture with LLMs

4.1 Role, Task and Agent Setup

CRAFT instantiates a **pragmatic speaker** role (Directors) and a **pragmatic listener** role (Builder), directly reflecting the multi-sender BPS framework (Sec. 3.1) and the well-established asymmetry between pragmatic production and comprehension [Ferreira, 2005, Sieker and Zarri  , 2026]: Directors must *produce* pragmatically calibrated instructions under partial observability—the harder and

³This framing is consistent with Gricean cooperativity [Grice, 1975]: the joint ToM listener's calibration requirement subsumes the Maxim of Quantity in a grounded multi-agent setting.

⁴In CRAFT, reachability is instantiated by oracle-verified moves—physically valid actions guaranteed to make forward progress from s_t toward $\mathcal{T}$ (Sec. 5.2).

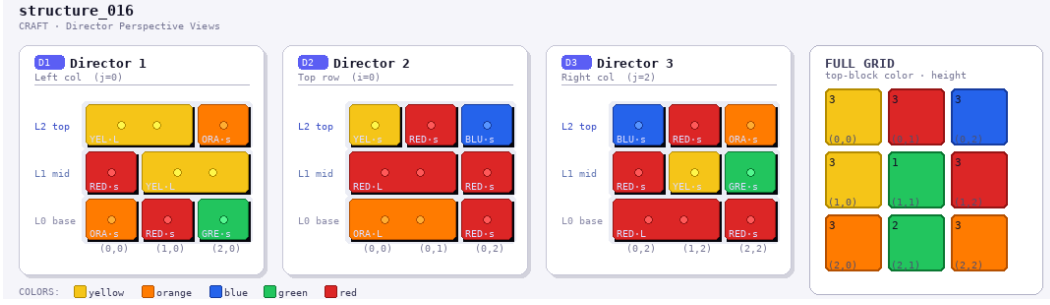


Figure 2: Director perspective views for a 25 block structure with the “camera” at the “bottom” of the *Full Grid* view. D1 (left wall), D2 (far wall), and D3 (right wall) each represent a fixed 2D projection across all vertical layers. The Full Grid minimap shows ground-truth stack heights.

less reliable capability in LLMs—while the Builder *interprets* and acts on those instructions, a task LLMs handle more reliably. This is reflected in the Director’s outputs which are full natural language utterances focusing on pragmatic production while the Builder has a restricted action space (e.g., selection of moves) focusing on pragmatic comprehension.

View Geometry As shown in Fig. 2, **three Director agents** (D1, D2 and D3) and **one Builder agent** collaborate in a “turn-by-turn” synchronous manner [Ivison et al., 2023, Nath et al., 2025b] to reconstruct the target 3D structure on a 3×3 grid, where each “cell” holds a stack of up to three small or large (*domino*) blocks in one of five colors. Each Director has a private 2D projection of the target corresponding to one wall of the structure (see Fig. 2), while the Builder observes only the current board state and the Directors’ natural language messages. Private views overlap only at corners shared by two Directors, so *no individual* can reconstruct the full target unilaterally, and the same physical block can appear as a different size to different Directors depending on whether both cells of a domino fall within their projection (cf. Fig. 2, D1 vs. D2). Success therefore requires each Director to reason about what information they uniquely hold and calibrate their public utterances accordingly (Sec. 3.1).

4.2 Director Agent

Each Director agent receives its private target view, the current board state, and the shared conversation history, and produces two outputs per turn: a private `<think>` block containing unconstrained spatial reasoning, and a public `<message>`. This two-part structure directly operationalizes the BPS base speaker and ToM listener distinction: the `<think>` block is where the director identifies missing blocks from its visible wall (S_{base_i}), and the public `<message>` is where that reasoning is distilled into an utterance calibrated given what the Builder and other Directors already know (L_{ToM_i}). Crucially, this separation allows CRAFT to diagnose whether a Director reasoned correctly but failed to transfer that into a sufficiently specified public message—a failure mode invisible to single-output architectures and directly measurable here through the gap between private reasoning quality and public communication quality. The Director prompt and its constituent parts are given in Figs. 9–11 in Appendix D.

4.3 Builder Agent

The Builder agent observes Director messages and conversation history each turn and executes **one** block placement or removal, or makes a request for clarification. It has no access to $\mathcal{T}$ and, by design, no independent goal toward z^* —its sole task is to interpret Director descriptions as a pragmatic listener. However, in this multi-agent LLM setting, the Builder’s own reasoning capacity introduces potential variance independent of Director communication quality. Oracle-verified candidates grounded in $\mathcal{T}$ address this by providing the Builder with pre-computed forward-progress moves from s_t , removing the need for independent reasoning about z^* : the Builder selects among verified candidates rather than inferring reachability itself, directly operationalizing the reachability condition under which communication failure is defined (Sec. 3.1). Since oracle moves reflect *locally* verified progress from s_t rather than a *globally* optimal trajectory toward $\mathcal{T}$, Directors must still coordinate coherently across turns for task completion to accumulate. Figs. 12–14 in Appendix D show the full Builder prompts. Appendix B provides experimental details on oracle candidates.

5 Experiments

5.1 Generating Target 3D Structures

Target structures are generated by first assigning stack heights to grid cells. There are seven *required* cells around the edges of the board, excluding the “bottom edge” (Fig. 2), which have 3 layers of blocks. The remaining 2 cells receive a uniformly sampled height 0 – 2. Each layer is then tiled according to the calculated height at each cell using mix of small or large blocks. A large block occupies two adjacent cells *on the same layer*, forming a *domino* pair. Small blocks occupy a single position. Colors are sampled uniformly from the five options. Appendix B.3 provides details.

5.2 Collaborative Construction Gameplay

We evaluate LLMs acting as the Directors and the Builder building 20 structures drawn from our evaluation set, spanning 7 simple, 8 medium, and 5 complex configurations with block counts ranging from 21 to 25 (mean 23.2). Every game starts from an empty board. Each Director is assigned a personality archetype [Jung and Hull, 2023] or “persona” [Sun et al., 2024]. This is done to meaningfully increase diversity in Director utterances so that they do not converge to similar templatic statements, and are assigned deterministically to ensure consistent roles across all model evaluations. A game for each model-structure combination consists of 20 communicative turns, each of which consists of collecting responses from a randomized selection of one to three unique Directors sequentially, followed by a Builder move selection conditioned on the Director discussion that updates the current board (See Fig. 1). This reflects natural multi-party conversation dynamics [Ganesh et al., 2023], where Director participation varies per turn—on average two Directors contribute per turn. Additionally, in each turn, the current conversation history of all prior Director responses is shown to the Directors, while the Builder only observes the current turn’s Director responses. For details on structures and Director personalities, see Table 2 in Appendix B.

5.3 Evaluation Strategy

We report metrics at turn 20 that measure how close the final board state is to the target structure. The most important are **completion percentage**, or percentage of blocks whose color and position exactly match those of the final structure; and **overall progress**, the unweighted mean of completion percentage, IoU of the current vs. target structure, and accuracy of block *colors* in each cell independent of vertical layer order. We additionally report remove gap (Gap = REMOVE – oracle remove rate), averaged across turns. A positive gap indicates Directors instructed more removals than the board required. Appendix B.5 provides more details on task metrics.

Automatic Grading A pragmatically competent Director produces a message that is not a transcription of its internal reasoning but a selective, non-redundant instruction calibrated to current shared knowledge. This motivates three diagnostically-independent *multidimensional* LLM “graders” [Zheng et al., 2023, Davidson et al., 2025].⁵ The **Spatial Grounding** (SG) judge evaluates whether each Director’s private <think> block correctly identifies the missing block, its layer, size, and physical executability (F1; prompt with questions in Fig. 16). The **Mind Modeling**⁶ (MM) judge evaluates whether the Director’s public message adds novel information, leverages the Director’s unique wall perspective, and acknowledges conflicts with other directors (F2; Fig. 17). The SG and MM judges measure the quality of a Director’s individual reasoning and public message, but a communication failure (Sec. 3.1) can still happen if the *collective* Director output was insufficient for a rational Builder to identify an oracle move. As such, the **Pragmatic Sufficiency** (PS) judge evaluates whether the collective Director output was sufficient for a rational Builder to identify at least one oracle-correct move—the empirical instantiation of $L_{\text{ToM}_{\text{joint}}}$ placing sufficient mass on z^* (Sec. 3.1; Fig. 18), a property irreducible to any individual Director’s utterance.

We score each grader question by mapping the model response to a value: *Yes* = 1, *No* = 0, and *Unclear* = 0.5. Overall SG and MM scores are the mean across all questions for a given director turn; the PS score is the mean across applicable questions for a given collective turn. All scores are averaged across **three** independent grader runs to reduce variance from stochastic model outputs. All judge prompts with questions are provided in Figs. 16–18 (Appendix E).

⁵Due to the large scale experiments in our evaluations involving thousands of trajectory logs with long prompts, human evaluation was not feasible.

⁶We use “Mind Model” in the sense of evaluating $L_{\text{ToM}_{\text{joint}}}$ qualities in Director messages [Riemer et al., 2024, Xu et al., 2024], not as a claim about functional ToM capacities [Mahowald et al., 2024].

Model	Prog.↑	Comp.↑	Pos. Acc.↑	IoU↑	Fail↓	REMOVE↓	Gap↓
Gemini-3-Flash	0.675 ± 0.052	0.716 ± 0.045	0.594 ± 0.065	0.817 ± 0.034	0.625 ± 0.125	0.196	0.018
GPT-4o	0.588 ± 0.051	0.633 ± 0.048	0.500 ± 0.061	0.753 ± 0.038	0.421 ± 0.116	0.280	0.056
GPT-4o-Mini	0.333 ± 0.041	0.383 ± 0.040	<u>0.233</u> ± 0.047	0.531 ± 0.041	0.550 ± 0.114	0.432	0.254
GPT-4.1-Mini	0.312 ± 0.053	0.352 ± 0.054	<u>0.233</u> ± 0.056	0.481 ± 0.053	0.500 ± 0.115	0.463	0.388
Claude-Sonnet-4.6	0.285 ± 0.036	0.332 ± 0.038	0.189 ± 0.039	0.479 ± 0.041	0.350 ± 0.109	0.395	0.265
Gemini-2.5-Flash	<u>0.257</u> ± 0.030	0.279 ± 0.033	0.206 ± 0.033	0.428 ± 0.039	0.300 ± 0.105	0.467	0.402
Gemini-3.1-Flash-lite	0.257 ± 0.052	<u>0.273</u> ± 0.055	0.286 ± 0.054	<u>0.211</u> ± 0.0514	0.550 ± 0.114	<u>0.540</u>	<u>0.467</u>
Mistral-7B	0.631 ± 0.053	0.673 ± 0.046	0.539 ± 0.067	0.793 ± 0.033	0.500 ± 0.129	0.124	-0.124
Qwen-7B	0.612 ± 0.044	0.665 ± 0.040	0.517 ± 0.052	0.778 ± 0.031	0.556 ± 0.121	0.205	-0.116
Llama-8B	0.586 ± 0.052	0.630 ± 0.049	0.506 ± 0.056	0.741 ± 0.057	0.684 ± 0.110	0.277	0.080
Gemma-9B	0.578 ± 0.044	0.628 ± 0.039	0.483 ± 0.054	0.751 ± 0.033	0.600 ± 0.112	0.122	-0.084
Qwen-72B	0.557 ± 0.049	0.606 ± 0.042	0.461 ± 0.064	0.733 ± 0.036	0.421 ± 0.116	0.245	0.047
Qwen-14B	0.476 ± 0.063	0.514 ± 0.061	0.394 ± 0.068	0.642 ± 0.055	0.611 ± 0.118	0.355	0.229
DeepSeek-Lite	0.419 ± 0.041	0.474 ± 0.039	0.317 ± 0.048	0.617 ± 0.038	0.400 ± 0.112	0.148	-0.275
Qwen-32B	0.339 ± 0.045	0.378 ± 0.048	0.250 ± 0.046	0.530 ± 0.048	<u>0.850</u> ± 0.082	0.448	<u>0.374</u>

Table 1: Task performance in CRAFT across frontier models (top) and open-weight models (bottom) across 20 games after turn $t = 20$. The Builder receives up to 5 “legal” moves per turn and selects among them based solely on Director instructions. *Prog.*, *Comp.*, *Pos. Acc.*, and *IoU* are continuous metrics; *Failed* is a binary per-turn outcome. Subscripted values show Mean $\pm$ SEM. REMOVE and Gap are aggregated over all turns. **Bold** = best within group. Underline = worst across all models.

Models and Implementation We evaluate 15 Director models. Eight are open-weight 7B–72B parameter models: Qwen-2.5-Instruct 7B/14B/32B/72B [Qwen et al., 2025], Llama-3.1-8B Instruct [AI@Meta, 2024], Mistral-7B-Instruct-v0.3 [Jiang et al., 2023], Gemma 2 9B Instruct [Team et al., 2024], DeepSeek-V2-Lite [Shao et al., 2024]. Seven are frontier/proprietary models: GPT-4o, GPT-4o-Mini, GPT-4.1-Mini [OpenAI, 2023b], Claude-Sonnet-4.6, Gemini-2.5-Flash, Gemini-3-Flash, Gemini-3.1-Flash-Lite.⁷ All are paired with a fixed GPT-4o-mini [OpenAI et al., 2024] Builder for all experiments. Each model is evaluated 3 times independently on 20 test structures over 20 turns. Open-weight models use a 512-token output budget; frontier models use 2,000 tokens (GPT series) or 3,000 tokens (Claude, Gemini) to accommodate extended chain-of-thought generation without truncating reasoning traces. For the LLM-grading evaluation (Sec. 5.3), we use GPT-4o-mini. Appendix G contains additional notes on experimental configurations.

6 Results

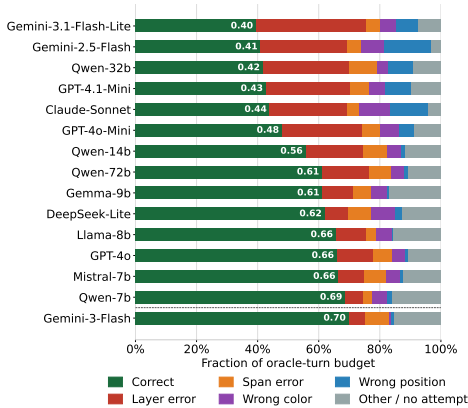


Figure 3: Failure taxonomy over all turns across 15 director models.

Wrong position errors (blue) indicate that Directors like Claude-Sonnet and the two lowest-performing Gemini models likely issue instructions that map to incorrect board locations. Span errors (orange) are

Frontier models do not uniformly dominate open-weight models Table 1 shows substantial variation within both groups. Gemini-3-Flash leads all 15 models with average progress of 0.675, but GPT-4.1-Mini (0.312), Claude-Sonnet-4.6 (0.285), and both other Gemini-Flash variants (0.257) fall below most open-weight models. This shows that perspective-based spatial reasoning can be challenging even for frontier models, consistent with prior work [Li et al., 2025].

Move-Level Failures Fig. 3 shows a taxonomy of failures across all 15 models, computed by replaying saved game logs and classifying each turn in which oracle-verified moves were available⁸ by its dominant failure mode. Layer errors (red) dominate across nearly all models, confirming that 3D layer inference is the primary move-level bottleneck—Directors likely correctly identify target positions but specify a vertical layer where the move is invalid and fails.

⁷For brevity, we refer to models by shortened names hereafter (e.g., Qwen-7B for Qwen-2.5-Instruct-7B, Mistral-7B for Mistral-7B-Instruct-v0.3, Gemini-3-Flash for Gemini-3-Flash-Preview).

⁸Turns without valid oracle moves rarely exceed 15% of total turns across all models on average. Figure 7 provides details.

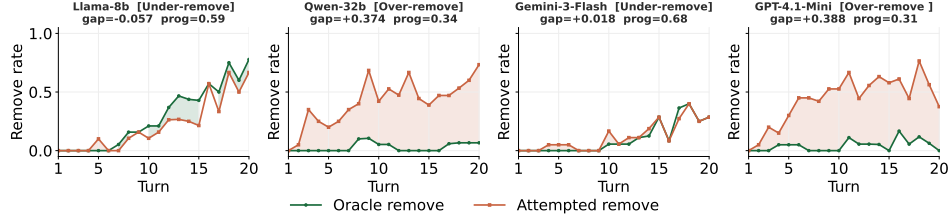


Figure 4: Per turn oracle-prescribed vs. attempted remove rate, averaged across all 20 structures (shading = *remove gap*, or difference between fraction of turns where Directors instruct a remove action and fraction where oracle prescribes one; **green**: model under-removes, **orange**: model over-removes). Subplot titles show mean gap and final-turn task progress.

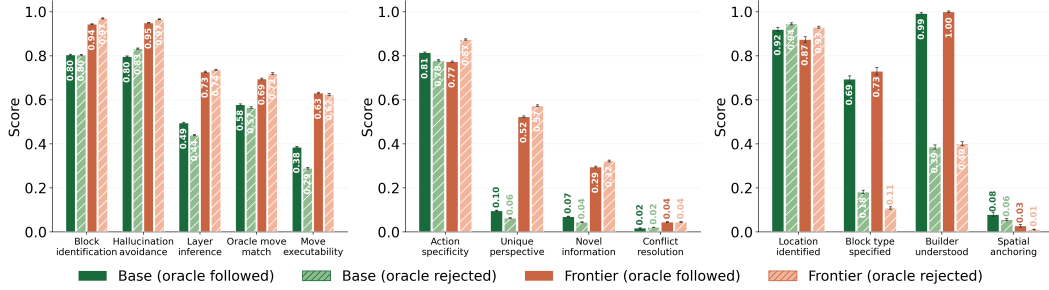


Figure 5: LLM grader scores across **three** evaluation dimensions—**spatial grounding** (left), **mind modeling** (center), and **pragmatic sufficiency** (right)—broken down by question and model group. Error bars denote standard error of the mean across all structure–turn–director observations per model from 3 independent runs.

elevated for Gemini-3-Flash relative to its otherwise strong performance: despite Directors correctly identifying block type and position, the Builder fails to infer the required second-cell endpoint for large blocks from the Director instructions. This is compounded by the partial observability design—a large block spanning two cells appears as size 2 to the Director whose view contains both cells, but as size 1 to Directors who see only one face, making it difficult for Directors to consistently communicate domino placement with sufficient precision for the Builder to resolve the span (see Appendix B.2 for details on block encoding and partial view projection).

Correction spirals as a signature of communication failure The *remove gap*, or the difference between the fraction of turns where Directors instruct a remove action and the fraction where the oracle prescribes one, separates frontier from open-weight models most clearly and is strongly negatively correlated with oracle adherence ($r = -0.543$, $p < 0.001$), and therefore task progress. Fig. 4 plots the *evolution* of builder remove actions across turns against oracle-prescribed remove actions across all evaluations.⁹ While 95.4% of oracle-prescribed moves are placements, the fraction of turns requiring a removal grows sharply after turn 10 as boards accumulate errors. Low-performing models show over-removals early in the game before any board errors have accumulated—indicating that over-removal is a proactive Director communication pattern rather than a reactive response to mistakes. This “correction spiral” reflects a fundamental property of the multi-perspective coordination requirement: three Directors may simultaneously identify different blocks to remove, leaving the Builder to resolve conflicting priorities across turns without guaranteed progress. In contrast, well-performing models (Llama-8B, Gemini-3-Flash) track the oracle remove signal closely throughout the game.

6.1 LLM Grader Results

Oracle adherence and remove behavior characterize *what* goes wrong and *how often*, but do not explain *why*. We therefore further investigate communication failures with LLM graders.

Individual reasoning and message quality do not distinguish turn or task success Fig. 5 shows automatic grading results¹⁰ for **spatial grounding** (left), **mind modeling** (middle), and **pragmatic**

⁹Full per-model evolution plots for all base and frontier models appear in Figs. 20 and 21 in Appendix F.

¹⁰Detailed results are shown in Table 7 and Fig. 19 in the Appendix.

sufficiency (right) of Director internal reasoning and public messages, grouped by Builder oracle adherence. This reveals a key turn-level diagnostic: SG and MM scores are nearly identical given oracle adherence condition, indicating that Directors reason and communicate at similar quality regardless of whether the Builder ultimately follows the oracle. In contrast, PS scores drop sharply when the Builder does not execute the oracle-recommended move. This suggests that individual communication quality (SG and MM scores) does not distinguish successful from unsuccessful turns, whereas collective sufficiency does—directly operationalizing the group-level irreducibility established in Sec. 3.1. This is expected: the PS judge evaluates the joint Director output as a whole, which is precisely the object whose sufficiency determines whether the Builder can identify a correct move, regardless of any individual Director’s communication quality.

Frontier models score higher on spatial grounding (0.829 ± 0.036 vs. 0.658 ± 0.069) and mind modeling (0.642 ± 0.067 vs. 0.502 ± 0.024), yet achieve lower task progress (0.387 vs. 0.525). Better individual reasoning and message quality, in other words, does not automatically produce better task outcomes. This calls into question the utility of using benchmarks of individual communication for multi-agent tasks; higher-level qualities of the information exchange are more explanatory. Table 7 in Appendix E reports per-model judge scores with standard errors.

Individual Director Communication vs. Collaborative Task Performance We correlated all judge dimensions with progress across all Directors.¹¹ Most show negative correlations with progress, but **unique perspective utilization (MM5)** stands out: it directly measures whether a Director reasons about what is “exclusively visible” from their wall, which is precisely what would lead a Director to identify board corrections invisible to others and instruct removals accordingly. Mediation analysis [MacKinnon et al., 2007] confirms the likely causal chain. Unique perspective utilization alone explains $R^2 = 0.330$ of progress variance, remove gap alone explains $R^2 = 0.609$, and both together explain $R^2 = 0.633$, a gain of only 0.024. Controlling for remove gap establishes full mediation: the partial correlation of unique perspective with progress is non-significant ($r = -0.247$, $p = 0.374$). **Directors who carefully leverage their private wall view produce more correction-oriented instructions, which drive over-removal, which halts progress over the 20 turns.** This suggests that the very sophistication that makes frontier Directors better individual communicators makes them relatively worse collective coordinators in CRAFT’s multi-sender setting.

7 Conclusion

We introduced CRAFT, a multi-agent benchmark for coordination under asymmetric information, grounded in a multi-sender Bounded Pragmatic Speaker (BPS) formulation in which optimal behavior requires calibrating to a *joint* listener. Empirically, we find a clear dissociation between individual communicative competence and collective task success: frontier models score higher on communication metrics yet underperform smaller open-weight models on task completion. This gap reflects a frontier-specific failure mode consistent with our theory—Directors that better leverage their unique perspective produce correction-heavy instructions that satisfy individual ToM objectives but misalign with the joint listener, inducing over-removal and exhausting the turn budget. Open-weight models, despite weaker individual competence, avoid this failure due to less aggressive epistemic updating. No model successfully models the joint listener in practice. These results show that individual pragmatic competence is necessary but not sufficient for coordination, and that failure to approximate the joint listener leads to systematic breakdowns. More broadly, CRAFT demonstrates that benchmarks of individual pragmatic reasoning are insufficient for evaluating multi-agent systems: coordination and individual competence are empirically dissociable and can move in opposite directions.

Limitations and Future Work CRAFT is evaluated in a *text-only* setting and does not consider models with direct access to visual inputs [Li et al., 2025, Liu et al., 2025]. While multimodal inputs may improve individual spatial grounding, it remains unclear whether such gains transfer to multi-agent coordination. Our experiments also use a fixed Builder with access to oracle-assisted candidate moves. This controlled design isolates the effect of Director communication and enables clearer credit assignment, but does not fully reflect real-world settings where agents must act without access to ground-truth guidance or may need to “explore” possible moves. We also do not explore settings with heterogeneous Director models (e.g., mixing open and proprietary models within a game) [Davidson et al., 2025]. While we empirically validated the credit attributed to each Director in the *joint listener* for the Builder (Eq. 3) for CRAFT games in Appendix F.1, heterogeneous agents in the loop would

¹¹Complete results shown in Table 8 in Appendix E and E.2.

naturally require more sophisticated methods of credit assignment from MARL research such as counterfactual credits [Deshmukh et al., 2026], or more principled approaches like Shapley-Owen values [Shapley, 1953, Nath et al., 2026]. Such configurations could provide further insight into how differences in alignment algorithms, pretraining, and post-training data influence collaborative performance and agent behavior, including partner-aware coordination in information-asymmetric settings [Curvo, 2025, Liang et al., 2025, Hu et al., 2021, Nath and Krishnaswamy, 2025]. While our controlled setup reduces variability and improves interpretability, it limits conclusions about joint optimization in heterogeneous multi-agent systems, where communication and execution policies may co-adapt.

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A Proofs

Proof. We proceed in three steps in order to prove Theorem 1.

Step 1: Per-Director optimization. Each Director D_i selects their utterance u_i to maximize expected utility under the joint listener. Since the joint listener $L_{\text{ToM}_{\text{joint}}}$ evaluates the full tuple (u_1, u_2, u_3) , the optimal utterance for D_i solves:

$$u_i^* = \arg \max_{u_i \in \mathcal{U}} \log S_{\text{base}_i}(u_i | z_i^*, c_i) + \log L_{\text{ToM}_{\text{joint}}}(z^* | u_1, u_2, u_3, c), \quad (5)$$

where we work in log space for convenience. This is the standard BPS objective (Eq. 1) applied to D_i , with $L_{\text{ToM}_{\text{joint}}}$ playing the role of the ToM listener.

Step 2: Conditional independence of base speakers. Each base speaker S_{base_i} is parameterized independently and therefore it conditions on D_i 's private context $c_i = (o_{i,t}, h_t, \{u_{j,t}\}_{j \neq i})$ and intention z_i^* , and does not depend on the utterances or intentions of the other directors except through the shared conversation history h_t already included in c_i . Formally, for $i \neq j$:

$$S_{\text{base}_i}(u_i | z_i^*, c_i) \perp\!\!\!\perp S_{\text{base}_j}(u_j | z_j^*, c_j) \Big| c, \quad (6)$$

where $c = (h_t, s_t)$ is the shared public context. Under this conditional independence, the joint distribution over all three utterances given the shared context factorizes as:

$$p(u_1, u_2, u_3 | z^*, c) = \prod_{i=1}^3 S_{\text{base}_i}(u_i | z_i^*, c_i). \quad (7)$$

Step 3: Deriving the joint policy. Combining Eq. 7 with the joint listener $L_{\text{ToM}_{\text{joint}}}$ (Eq. 3), the optimal joint policy is obtained by multiplying the factorized base speaker distribution by the joint listener and normalizing:

$$\begin{aligned} \pi^*(u_1, u_2, u_3 | z^*, c) &\propto p(u_1, u_2, u_3 | z^*, c) \cdot L_{\text{ToM}_{\text{joint}}}(z^* | u_1, u_2, u_3, c) \\ &= \left(\prod_{i=1}^3 S_{\text{base}_i}(u_i | z_i^*, c_i) \right) \cdot L_{\text{ToM}_{\text{joint}}}(z^* | u_1, u_2, u_3, c). \end{aligned} \quad (8)$$

This is exactly Eq. 4, completing the derivation.

To verify that Eq 8 preserves the BPS form, substitute Eq. 3 for $L_{\text{ToM}_{\text{joint}}}$:

$$\begin{aligned} \pi^*(u_1, u_2, u_3 | z^*, c) &\propto \left(\prod_{i=1}^3 S_{\text{base}_i}(u_i | z_i^*, c_i) \right) \cdot \exp \left(\sum_{i=1}^3 \lambda_i R_i(u_i, s_t, \mathcal{T}) \right) \\ &= \prod_{i=1}^3 [S_{\text{base}_i}(u_i | z_i^*, c_i) \cdot \exp(\lambda_i R_i(u_i, s_t, \mathcal{T}))], \end{aligned} \quad (9)$$

where the last step uses the fact that the exponential of a sum factorizes into a product of exponentials. Each factor in Eq. 9 has exactly the BPS form of Eq. 1, with S_{base_i} as the base speaker and $\exp(\lambda_i R_i/Z_i)$ as the per-director ToM listener, confirming that the multi-sender joint policy is a product of individual BPS policies. $\square$

Lemma 1 (Single-Sender BPS as Special Case). *Theorem 1 reduces to the standard single-sender BPS (Definition 1) under $N = 1$.*

Proof. In order to show that the single-sender BPS framework is a special case of Theorem 1, we consider the setup where $N = 1$. The private context of D_1 reduces to $c_1 = (o_{1,t}, h_t, \emptyset)$ since $\{u_{j,t}\}_{j \neq 1} = \emptyset$, and $z^* = z_1^*$. The joint ToM listener (Definition 2) reduces to:

$$L_{\text{ToM}_{\text{joint}}}(z_1^* | u_1, c_1) \propto \exp(\lambda_1 R_1(u_1, s_t, \mathcal{T})) =: L_{\text{ToM}_1}(z_1^* | u_1, c_1), \quad (10)$$

since all terms $\lambda_i R_i = 0$ for $i > 1$. Substituting into Equation 4, the product over base speakers collapses to a single factor:

$$\pi^*(u_1 | z_1^*, c_1) \propto S_{\text{base}_1}(u_1 | z_1^*, c_1) \cdot L_{\text{ToM}_1}(z_1^* | u_1, c_1), \quad (11)$$

which is Equation 1 exactly. $\square$

Lemma 1 suggests that the single-sender approximation is insufficient whenever Directors hold complementary information. In Figure 2, no single Director observes the full structure: D1 sees the left wall, D2 the top row, D3 the right wall, with overlap only at shared corners. Communicative sufficiency for the Builder therefore requires that the *collective* output (u_1, u_2, u_3) covers what no individual Director can specify alone—a condition invisible to any single-sender evaluation and only measurable at the group level.

Connection Between Homogeneity of Director Groups and Equal Weighting in joint ToM listener (via λ_i in Eq. 3). In general, the weighting coefficients λ_i in the joint ToM listener for the Builder agent capture how the Builder aggregates per-Director contributions and may reflect differences in capability, information asymmetry, or task geometry. In heterogeneous settings [Davidson et al., 2025], this can induce asymmetric weighting of these coefficients. In contrast, the homogeneous Director configuration evaluated in CRAFT is “permutation-invariant”: all Directors share the same underlying model and differ only in role and partial observation, and no viewpoint is structurally privileged. Under this symmetry, a consistent aggregation mechanism should assign equal weight to each Director’s contribution—a quality we empirically validate in Section F.1.

B Oracle Settings, Structure Generation and Director View Computation

B.1 Oracle Evaluation

Implementation The oracle condition operationalizes the reachability requirement established in Sec. 3.1: at each turn, we enumerate all moves that make verified forward progress from s_t toward $\mathcal{T}$, covering three cases: (i) *placement*—if the current stack is shorter than the target, the next required block (including color, size, and for large blocks, the span partner cell) is generated as a candidate; (ii) *excess removal*—if the current stack exceeds the target depth, a remove move for the topmost block is generated; (iii) *wrong block correction*—if the stacks are the same depth but a block at some layer is incorrect, a remove move is generated for the topmost block if it is the wrong one, or for the correct block above it if buried. All candidates are verified by simulating execution against a copy of the current game state before inclusion. Only moves that succeed in simulation and produce correct structural placement are retained. **Up to $N=5$ verified candidates** are sampled per turn as the Builder’s input observation for the main experiments reported here (Section 6) using a deterministic seed derived from the structure index and turn number, ensuring reproducibility across model comparisons. Large block candidates include the span partner cell explicitly, since both endpoints must be specified for valid execution.

Builder selection criterion Verified candidates are injected into the Builder’s prompt alongside the Director discussion, presented in a lightweight natural language format—for example, `PLACE gs @ (0,1) layer 0` for a small block or `PLACE bl @ (1,0) layer 0 → (2,0)` for a large block spanning two cells. The Builder is instructed to select the candidate that it believes at least one Director is describing, based on the current turn’s Director discussion. If no candidate clearly matches, the Builder issues a clarification request rather than selecting arbitrarily. This preserves the full pragmatic inference requirement: the Builder must still interpret Director natural language and map it to a specific candidate, consistent with its role as a neutral passive listener established in Sec. 4.3. The Builder’s confirmation field is extended to include a brief rationale identifying which Director(s) were followed and whether Directors agreed or conflicted, providing signal for subsequent analysis.

Oracle as a local progress signal The oracle verifies forward progress from s_t at each turn but has no lookahead: a locally correct move is not necessarily a strategically good one if it creates correction debt downstream. Whether the Builder follows oracle depends on whether Directors made the right candidate identifiable; whether locally correct moves accumulate into global completion depends on whether Directors maintained a coherent group strategy across turns rather than consuming the turn budget on redundant or misdirected instructions. Oracle adherence is therefore necessary but not sufficient for task completion—and the inversion we observe (Sec. 6), where frontier models with higher individual communication quality achieve lower task progress despite high oracle availability (mean 0.83–0.92 across all models; Fig. 7), is the empirical consequence of this: the oracle does not make the task easy, it makes Director group coordination the binding constraint. Oracle adherence

strongly predicts task progress ($r = 0.962$, $p < 0.001$), confirming reachability as the operative measure of communication quality, while no-oracle turns remain rare and uniform across model families, reflecting irrecoverable board states rather than Director failures.

B.2 Block Encoding and the World State

The world is represented as a 3×3 grid of positions, each identified by a coordinate pair (i, j) where $i, j \in \{0, 1, 2\}$. Each position holds an ordered stack of blocks, where the stack index corresponds to vertical layer. A block is encoded as a two-character string where the first character denotes color—green, blue, red, yellow, or orange—and the second denotes size: small (s) or large (l). The full set of valid block types is:

$$\mathcal{B} = \{gs, gl, bs, bl, rs, rl, ys, yl, os, ol\}$$

The world state at any point is a function $S : \mathcal{C} \rightarrow \mathcal{B}^*$, mapping each coordinate $c \in \mathcal{C} = \{(i, j) \mid i, j \in \{0, 1, 2\}\}$ to an ordered sequence of blocks (possibly empty), with stacks capped at height 3.

B.3 Structure Generation

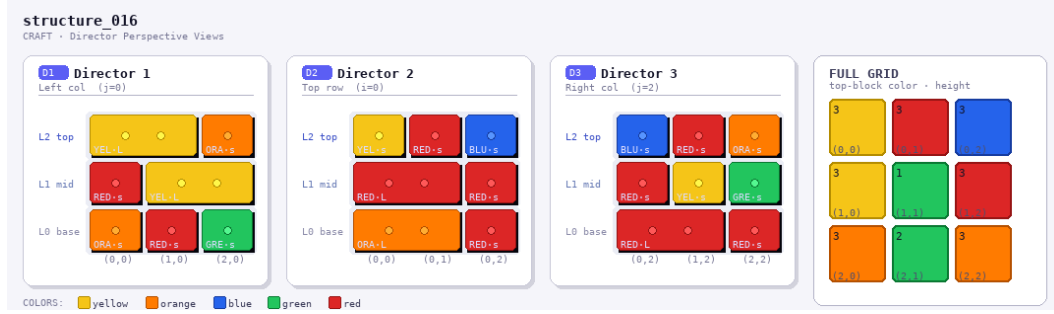


Figure 6: Director perspective views for **structure_016**, a complex-tier structure with 25 total blocks. **D1** (left column, $j=0$) sees a large yellow domino spanning $(0, 0) - (1, 0)$ at L2, a small red at $(0, 0)$ and large yellow domino spanning $(1, 0) - (2, 0)$ at L1, and small orange, red, green blocks at L0. **D2** (top row, $i=0$) sees a large orange domino spanning $(0, 0) - (0, 1)$ with small red at $(0, 2)$ at L0, a large red domino spanning $(0, 0) - (0, 1)$ with small red at $(0, 2)$ at L1, and small yellow, red, blue at L2. **D3** (right column, $j=2$) sees small blue, red, orange at L2, small red, yellow, green at L1, and a large red domino spanning $(0, 2) - (1, 2)$ with small red at $(2, 2)$ at L0. The shared position $(0, 0)$ is visible to both D1 and D2, serving as the sole grounding anchor between views. The **Full Grid** minimap shows all seven required positions at height 3; the optional positions $(1, 1)$ and $(2, 1)$ have heights 1 and 2 respectively, both topped by green blocks. No Director can reconstruct the full structure unilaterally: D1 and D3 observe vertical depth along their respective walls but cannot see interior or opposite-wall positions, while D2 has exclusive visibility into interior positions but cannot observe vertical structure below the topmost block. Notably, the large orange domino spanning $(0, 0) - (0, 1)$ at L0 is seen as a full domino by D2 (both cells visible) but appears as a small block to D1, which can only see $(0, 0)$ and has no visibility into $(0, 1)$. Similarly, the large red domino spanning $(0, 2) - (1, 2)$ at L0 is seen as a full domino by D3 but appears small to D2, which sees $(0, 2)$ but not $(1, 2)$. **These cases illustrate how the same physical block can have different apparent sizes depending on the observing Director’s projection, a direct consequence of the partial observability design.**

Target structures are generated by a two-stage process: first assigning stack heights to grid positions, then tiling each vertical layer independently with blocks.

Stack Height Assignment Grid positions are partitioned into two sets. Seven *required* positions—all positions except $(1, 1)$ and $(2, 1)$ —always receive exactly three layers of blocks. The two *optional* positions, $(1, 1)$ and $(2, 1)$, receive a height sampled uniformly from $\{0, 1, 2\}$, independently. This design ensures a dense, consistently tall structure at the periphery of the grid while allowing variable interior depth.

Layer Tiling Each layer is tiled independently. For a given layer, the set of positions that require a block at that depth is determined by the height assignments above. These positions are then filled using a mix of small blocks and large blocks. A large block occupies two orthogonally adjacent positions *on the same layer*—forming a domino pair—and is never stacked vertically. Small blocks occupy a single position. For each position, the generator probabilistically attempts to form a domino with an available orthogonal neighbor; if no neighbor is free or the attempt fails, a small block is placed instead. Colors are sampled uniformly from the five available colors. To discourage structurally repetitive configurations, the generator makes a small number of retry attempts to avoid assigning the same block type to the same position on consecutive layers.

Complexity Classification Structures are labeled post-hoc by total block count. Structures with at most 22 blocks are labeled *simple*, those with 23–24 blocks are labeled *medium*, and those exceeding 24 blocks are labeled *complex*. Because required positions always contribute 21 blocks (seven positions at three layers each), complexity variation is driven almost entirely by the optional positions and the proportion of large blocks, which can increase the count when domino pairs span positions that would otherwise be unfilled.

B.4 Director View Projections

Each Director agent is assigned a fixed 2D projection of the 3D world state, capturing a different face of the grid. The three projections are:

D1: Left Column View D1 observes positions $(0, 0)$, $(1, 0)$, and $(2, 0)$ across all three vertical layers, corresponding to the left-facing wall of the structure.

D2: Top Row View D2 observes positions $(0, 0)$, $(0, 1)$, and $(0, 2)$ across all three vertical layers, corresponding to the far-facing wall.

D3: Right Column View D3 observes positions $(0, 2)$, $(1, 2)$, and $(2, 2)$ across all three vertical layers, corresponding to the right-facing wall.

In each view, blocks are presented left-to-right according to the implied physical seating orientation of each Director (see [Zhu et al. \[2026\]](#)). Each cell in a view is encoded as a color–size pair; empty cells are represented as color none. A large block appears as size 2 only when both cells of its domino span fall within the Director’s visible positions; otherwise it appears as size 1, since only one face of the block is visible from that angle.

Information Coverage D1 and D3 share exactly one position, $(0, 0)$, providing a single grounding anchor between the two lateral views. D2 is the only Director with visibility into interior positions such as $(1, 1)$ and $(2, 1)$ —the optional positions—making D2 informationally pivotal for structures with non-trivial interior depth. No single Director can reconstruct the full 3D state unilaterally; productive coordination requires each Director to surface the information that the others structurally cannot observe.

B.5 Metrics

Progress toward the target structure S^* is measured after each successful move and computes four complementary metrics over the normalized representations of the current state S_t and target S^* .

Intersection over Union (IoU) For each position $c \in \mathcal{C}$, let $A_c = \{b \in S_t(c)\}$ and $B_c = \{b \in S^*(c)\}$ be the multisets of blocks treated as sets. The IoU score aggregates overlap across all positions:

$$\text{IoU}(S_t, S^*) = \frac{\sum_{c \in \mathcal{C}} |A_c \cap B_c|}{\sum_{c \in \mathcal{C}} |A_c \cup B_c|}$$

This metric is insensitive to block order within a stack and rewards partial position matches.

Archetype	Description	D1	D2	D3	Total
Assertive	Confident and direct; forms hypotheses quickly and shares them, updates when others provide compelling evidence.	75	60	30	165
Cautious	Methodical and verification-focused; synthesizes others' observations before adding interpretation.	60	90	75	225
Observant	Notices patterns and anomalies; flags inconsistencies and connects information across directors.	15	30	90	135
Skeptical	Questions assumptions including its own; probes claims to ensure group correctness, comfortable with uncertainty.	45	75	75	195
Synthesizer	Integrates all directors' observations into a coherent picture; reconciles contradictions and drives shared understanding.	105	45	30	180
Total		300	300	300	900

Table 2: Director personality archetypes used in CRAFT experiments. Each archetype shapes the Director’s internal reasoning style and public communication tone. Assignments are deterministic per (structure_index, run, director_id) ensuring consistency across all model evaluations.

Completion Percentage This metric measures layer-exact correctness—a block at position c and layer k counts as correct only if it matches $S^*(c)[k]$:

$$\text{CP}(S_t, S^*) = \frac{\sum_{c \in \mathcal{C}} \sum_{k=0}^{|S^*(c)|-1} \mathbf{1}[S_t(c)[k] = S^*(c)[k]]}{\sum_{c \in \mathcal{C}} |S^*(c)|}$$

Position Accuracy A coarser metric that rewards positions where the set of blocks matches exactly, regardless of layer order:

$$\text{PA}(S_t, S^*) = \frac{1}{9} \sum_{c \in \mathcal{C}} \mathbf{1}[\{b : b \in S_t(c)\} = \{b : b \in S^*(c)\}]$$

Overall Progress The scalar summary used for termination and trend analysis is the unweighted mean of the three metrics:

$$\text{OP}(S_t, S^*) = \frac{\text{IoU} + \text{CP} + \text{PA}}{3}$$

C Failure Taxonomy Details

To obtain the failure counts reported in the main paper, we replayed all saved game logs from the CRAFT evaluation and applied a deterministic taxonomy to every turn containing at least one oracle move. Each turn was first checked for a game engine error stored in `progress_data["error"]`; turns whose error message contained the substring “layer” or “span” were labeled *engine-layer* or *engine-span*, and all remaining engine failures were labeled *engine-other*.¹² For *engine-clean* turns (those with no game engine error), the attempted move was compared against the oracle set at three levels of strictness: a mismatch on action or position yielded *wrong-position*; a match on position but not block color yielded *wrong-color*; and a match on block but not span yielded *wrong-span*. Turns satisfying the full oracle match and accepted by the engine were counted as *correct*. Counts were normalized by the total number of oracle-labeled turns per model.

Oracle-based Outcome Breakdown Fig. 7 complements Table 3 by showing how the turn budget is allocated across all five outcome categories relative to total turns rather than oracle-available turns.

¹²Engine errors are not cases where the engine itself threw an unrecoverable error, but rather where the move failure stemmed from the engine rejecting a move for it being physically impossible given the current board context. The error signal serves as the signal of physical failure.

Type	Model	Correct	Layer	Span	Other	Wrong-Color	Wrong-Pos
Base	DeepSeek-Lite	.711	.086	.086	—	.089	.026
	Gemma-9b	.738	.120	.069	—	.066	.006
	Llama-8b	.774	.119	.036	.003	.062	.006
	Mistral-7b	.759	.094	.085	—	.053	.009
	Qwen-7b	.817	.070	.034	—	.061	.018
	Qwen-14b	.616	.209	.088	.025	.051	.011
	Qwen-32b	.453	.306	.100	.014	.038	.089
Frontier	Qwen-72b	.682	.170	.081	.003	.047	.017
	Gemini-3-Flash	.823	.061	.095	—	.009	.012
	GPT-4o	.734	.132	.070	.006	.045	.014
	GPT-4o-Mini	.513	.281	.064	.024	.067	.051
	Claude-Sonnet	.449	.258	.041	.020	.102	.130
	GPT-4.1-Mini	.457	.297	.064	.035	.056	.091
	Gemini-2.5-Flash	.412	.284	.048	.025	.075	.156
	Gemini-3.1-Flash-Lite	.401	.365	.046	.058	.056	.074

Table 3: Failure taxonomy for base and frontier models across CRAFT evaluation runs on 20 target structures. Each cell shows the fraction of oracle-available turns; **Correct** is the full oracle match rate. Engine errors are diagnosed from the game-engine response after the Builder agent makes a move; positional, color, and span errors reflect oracle-level mismatches.

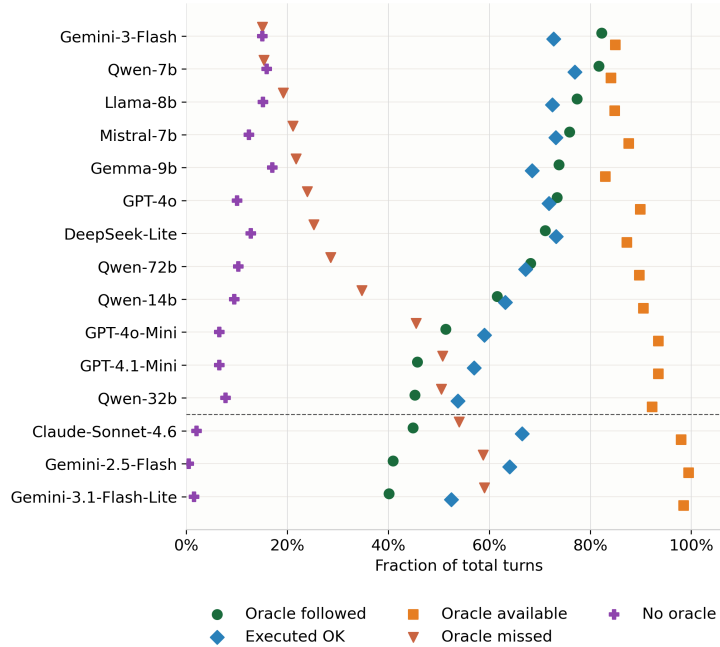


Figure 7: Turn-level outcome rates per model sorted by oracle adherence. Oracle available (orange) is uniformly high; the gap between oracle available and oracle followed (green) directly corresponds to the failure fractions in Table 3. No-oracle turns (purple) are rare across all models ($< 15\%$).

Oracle availability is uniformly high ($\geq 85\%$) across all models, confirming that the performance differences in the taxonomy table are not confounded by irrecoverable board states—the oracle budget was available to be followed in the vast majority of turns. The gap between available oracle moves and followed oracle moves in Fig. 7 directly corresponds to the *correct* bar in Table 3: models with small gaps (Gemini-3-Flash, Qwen-7b) show high correct rates in the table, while models with large gaps (Claude-Sonnet, Gemini-3.1-Flash-Lite) show low correct rates and high layer or wrong-position error fractions. The execution success rate (blue diamonds) diverges from oracle adherence most sharply for Claude-Sonnet-4.6 and Gemini-2.5-Flash, consistent with their elevated wrong-position

error rates in the taxonomy table—the Builder executes moves confidently but at incorrect locations as directed.

Time (s)	Δt (s)	Speaker	Utterance	Role
570.0	-29.6	Builder	This green?	Clarification query
571.1	-28.5	Director	Yes.	Confirmation
571.3	-28.3	Director	Yeah, stick that on top of the blue.	Instruction
572.7	-26.9	Builder	Right here?	Clarification query
572.9	-26.7	Director	No, no, no, no.	Correction
574.1	-25.5	Director	The yellow that we have, for me it is showing green.	Perceptual mismatch
577.6	-22.0	Director	It's not yellow.	Correction
579.3	-20.3	Builder	Cause, uh, is it like a, like a darkish green?	Hypothesis proposal
583.6	-16.0	Director	No.	Rejection
584.6	-15.0	Director	It's normal green.	Refinement
585.9	-13.7	Builder	I cannot see any yellow over there.	Perceptual conflict
587.2	-12.4	Builder	So this is green?	Clarification query
588.1	-11.5	Director	Yes.	Confirmation
588.4	-11.2	Director	That's yellow for me.	Misalignment signal
589.8	-9.8	Builder	This is yellow?	Clarification query
590.6	-8.9	Director	No.	Correction
591.1	-8.5	Director	The long.	Referential refinement
591.9	-7.7	Builder	This is yellow.	Hypothesis proposal
592.7	-6.9	Director	Yeah, that one's yellow.	Alignment confirmation
594.9	-4.7	Builder	So this is yellow.	Grounding confirmation
596.1	-3.5	Builder	Which one's this color?	Clarification query
597.3	-2.2	Builder	Green?	Hypothesis proposal
598.0	-1.6	Director	Yeah, the bottom one is green.	Final alignment
599.6	0.0	—	Action: REMOVE RS at LAYER 1	Execution

Table 4: Dialogue segment illustrating ambiguity resolution, perceptual misalignment, and multi-agent grounding prior to moving the small red block (RS) at layer 1. Δt is measured relative to the action at $t = 599.6s$ from Group 3.

C.1 Qualitative Error Analysis

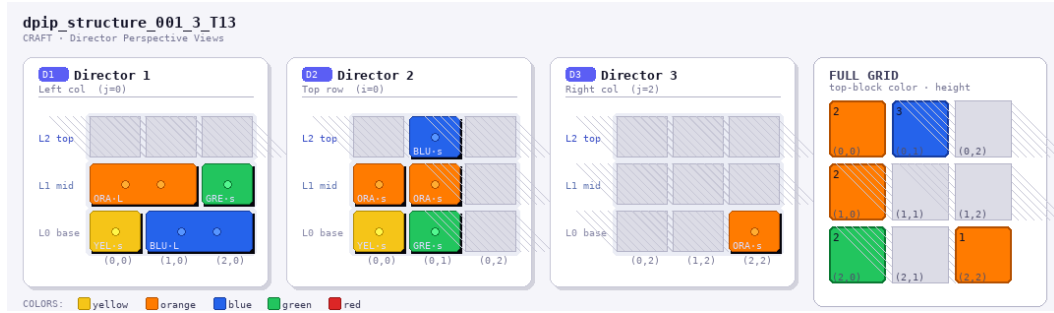


Figure 8: Three turns of zero progress in Qwen-32b (structure_001, T10–T14): D1 and D2 repeatedly instruct removal from the wrong layer while the oracle recommends forward placement, and the Builder follows Director consensus over oracle, consuming three turns without any board state change (full trace in Table 11).

We provide an example error analysis from the game runs for the Qwen-32b model as directors. Table 5 shows representative samples of errors from multiple open-weight and frontier-proprietary models, broken down by the Director model, turn of error, Director instructions for that turn and failure details. Fig. 8 shows the board state at turn 13 of the Qwen-32b correction spiral. Reading D1’s panel from bottom to top: layer 0 shows a small yellow block at (0,0) and a large blue domino spanning (1,0)–(2,0); layer 1 shows a large orange domino spanning (0,0)–(1,0)

Model	Turn	Director instruction	Failure & cause
F1 — Wrong block (base speaker failure)			
DeepSeek-Lite	T8	D1 & D3 (<i>identical</i>): “place a large orange block spanning (0,1) and (0,2)”	Builder places os at layer 2; oracle needs bs . Both directors describe target state not current state; redundant identical messages add zero information.
DeepSeek-Lite	T1	D2 & D3: “large orange block spanning middle and right of my bottom layer”	Builder places ol at (1,2) → (2,2); oracle needs ol at (0,0) span (0,1). Frame-of-reference ambiguity: “middle and right” resolves to wrong global coordinates.
F2 — Correction spiral (ToM listener failure)			
Qwen-32B	T13	D1: “remove the large orange block from the middle-left of my <i>bottom</i> layer”; D2: “remove the orange block from my <i>bottom</i> left corner”	Cannot remove layer 0 at (1,0) — ol sits at layer 1 not layer 0; both directors specify “bottom layer” without checking current stack depth. Oracle recommends place gs @ (0,0) layer 2 .
Qwen-32B	T14–15	D1 & D3: same remove instruction as T13	<i>Identical error, identical board state.</i> Three consecutive turns consumed; no director issues prerequisite remove (1,0) layer 1 needed to unblock target.
F3 — Span omission (reasoning–communication gap)			
Gemini-3-Flash	T10	D3: “put a <i>small</i> yellow block on my bottom right”	Builder places y1 without span_to ; oracle needs y1 at (1,0) span (2,0). Director said small; builder upgraded to large but omitted span endpoint.
Gemini-3-Flash	T6	D3 message truncated; D2 silent	Builder places bl at correct position but span_to=None ; oracle needs span (2,2). No span context in messages.
F3 — Layer miscounting (stacking constraint violation)			
Qwen-32B	T7	D2: “swap out the yellow block at bottom left”	Stack at (0,0): [ys , ol]. Builder attempts remove layer 0 ; must remove ol at layer 1 first. Director describes target without checking current stack depth.
Claude-Sonnet-4.6	T8	D1: “place small green at the near end of my left wall, <i>second level</i> ”	Stack depth at (0,0) is 2; correct layer is layer=2 but builder maps “second level” to layer=1 . Natural language level indexing misaligns with zero-indexed stack depth.
wrong value correct oracle value. Block codes: orange, blue, green, yellow, red.			

Table 5: Representative turn-level failures organized by BPS failure mode, as identified in Sec. 3.1. The Qwen-32b deadlock (Fig. 22) across turns 10–15 illustrates how a single wrong early placement traps directors in an irrecoverable correction spiral when no agent tracks the repair plan across turns.

and a small green at (2,0); layer 2 is entirely empty. The orange domino D1 wants to remove therefore sits at layer 1, directly accessible as the top block at both (0,0) and (1,0). However, D1’s instruction says “remove the large orange block from the middle-left of my *bottom* layer”—specifying layer 0, which is occupied by the blue domino, not the orange one. The game engine rejects the move with Cannot remove layer 0 at (1,0)—must remove top block first (layer 1). D2 makes the same error independently: it sees orange at layer 1 on (0,0) but instructs removal from the bottom corner, again specifying layer 0. D3 sees only a single orange block at (2,2) and has *no* visibility into either problem position. Critically, this board state and these Director instructions are *identical* across turns 13, 14, and 15—because neither the board nor the Directors’ instructions change after a failed remove attempt. Without any mechanism to track that the previous instruction failed and why, Directors re-issue the same “bottom layer” instruction each turn, consuming three turns without any forward progress while the oracle continues to recommend placement moves at other positions.

D Agent-Specific Prompts

We provide detailed prompts used in our experiments for both the Director (Figs. 9–11) and Builder agents (Figs. 12–14) in this section.

Director Prompt (I): Identity and Perspective

You are Director {D_i} in a collaborative LEGO construction task. You are sitting around a physical board with a Builder and two other Directors. From where the builder sits, D1 is to their left, D2 is across from them, and D3 is to their right.

YOU ARE {archetype}.

YOUR PERSONALITY:
{personality}

VERY IMPORTANT: You must adopt this personality in both your internal reasoning and your public utterances.

YOUR PERSPECTIVE:
From left to right, you see the following cells across all layers:
{perspective_description}

SPATIAL ORIENTATION (use only in your thinking)
The coordinate grid from above:
(0,0) (0,1) (0,2) ← this is the "far" / "back" row
(1,0) (1,1) (1,2)
(2,0) (2,1) (2,2) ← this is the "near" / "front" row
Large blocks span SIDEWAYS or FORWARD/BACK — never stacked vertically.

HOW TO INTERPRET YOUR TARGET VIEW
- IMPORTANT: In the JSON, keys are named row_0/row_1/row_2, but they refer to LAYERS (vertical stack depth), not grid rows.
- row_0 = layer_0 (bottom layer / stack depth 0)
- row_1 = layer_1 (middle layer / stack depth 1)
- row_2 = layer_2 (top layer / stack depth 2)
- in each layer, blocks are listed from LEFT to RIGHT according to YOUR VIEW
- In your PUBLIC message, say "bottom layer / middle layer / top layer" (avoid saying "bottom row").
- color=none means that cell should be empty
- size of 1 = the block is a small block, size of 2 = the block is a large block and spans two adjacent cells
- if two adjacent cells in your target view have the same color and BOTH are size 2, this means that a SINGLE large block occupies both those cells

EXAMPLE ANALYSIS OF TARGET VIEW AND BOARD STATE

D2's target view:

```
"D2": {
  "row_0": [
    { "color": "blue", "size": 1 },
    { "color": "orange", "size": 2 },
    { "color": "orange", "size": 2 }
  ],
  "row_1": [
    { "color": "yellow", "size": 1 },
    { "color": "yellow", "size": 1 },
    { "color": "orange", "size": 1 }
  ],
  "row_2": [
    { "color": "yellow", "size": 1 },
    { "color": "blue", "size": 1 },
    { "color": "green", "size": 1 }
  ]
}
```

Figure 9: Director prompt used in the experiments, section I. Note that personality and archetypes are populated from Table 2.

Director Prompt (II): Board State and Analysis

Current board state:

```
{ (0,0):[], (0,1):[], (0,2):[]; (1,0):[], (1,1):[], (1,2):[]; (2,0):[], (2,1):[], (2,2):[] }
```

Correct D2 analysis:

[From my perspective, the current board state has all cells empty. My target view specifies that layer 0 should have a blue small block in my bottom left corner (0,0), and then a large orange block spanning the middle and right cells (0,1) and (0,2).

Going left to right, layer 1 should have two small yellow blocks at (0,0) and (0,1), and a small orange block at (0,2).

Finally, layer 2 should consist of a yellow small block at (0,0), a blue small block at (0,1), and a green small block at (0,2).

To start, I need the builder to place a large orange block spanning (0,1) and (0,2), which are the middle and right cells of my bottom layer. This is the first action to align with my target view.]

Correct D2 utterance based on this analysis:

[Put a large orange block across the middle and the right side of my bottom layer.]

YOUR JOB:

Help the builder complete the structure by giving clear and correct instructions based on your private view.

RULES FOR REASONING (use only in your thinking):

- Carefully compare your target view with the current board state.
- Identify missing blocks, incorrect blocks, or incorrectly placed blocks.
- Determine the correct color, size, and layer for each required block.
- Respect physical constraints: blocks must be placed on valid support and large blocks must span correctly.
- Plan instructions that move the current board closer to your target view.
- Do not assume access to other directors' views; reason only from your own perspective.

Figure 10: Director prompt used in the experiments, section II.

Director Prompt (III): Speaking Rules, Examples, Inputs, and Response Format

RULES FOR SPEAKING (in your public message):

- Give clear, concise, and actionable instructions to the builder.
- Use natural language descriptions (e.g., "left", "right", "middle") based on your perspective.
- Refer to layers as "bottom", "middle", or "top" (not row numbers).
- Specify block color, size, and placement clearly.
- Avoid mentioning coordinates or JSON-style representations.
- Do not include internal reasoning in your message.
- Focus on one step or a small number of steps that the builder can execute reliably.

EXAMPLE UTTERANCES:

{examples}

CURRENT BOARD STATE:

{current_board_state}

TARGET VIEW:

{target_view}

CONVERSATION HISTORY:

{conversation_history}

RESPONSE FORMAT:

Return your response in the following format:

```
<analysis>
[Your internal reasoning here]
</analysis>
<message>
[Your instruction to the builder here]
</message>
```

Figure 11: Director prompt used in the experiments, section III.

Builder Prompt (I): Role, Perspective, Action Space and Environment Rules

You are a Builder in a collaborative LEGO construction task.

The three Directors (D1, D2, and D3) have to instruct you to build a single structure that is consistent with the private views of the structure they have.

Your job is to place, move, or remove blocks on the board to build the structure.

From a top-down view of the target structure, D1's private view is of the left wall of the structure, D2's view is of the top wall of the structure, and D3's view of the right wall of the structure.

From where the builder sits, D1 is to their left, D2 is across from them, and D3 is to their right.

SPATIAL ORIENTATION (use only in your thinking)

The coordinate grid from above:

(0,0) (0,1) (0,2) ← this is the "far" / "back" row

(1,0) (1,1) (1,2)

(2,0) (2,1) (2,2) ← this is the "near" / "front" row

Large blocks span SIDEWAYS or FORWARD/BACK — never stacked vertically.

DIRECTOR PERSPECTIVE GUIDE:

D1: From left to right, sees cells (0,0), (1,0), (2,0) across all layers.

D2: From left to right, sees cells (0,0), (0,1), (0,2) across all layers.

D3: From left to right, sees cells (0,2), (1,2), (2,2) across all layers.

When interpreting the instructions from D1, D2, or D3, you MUST adopt the frame of reference of the speaker.

For instance, to D1, "my bottom left corner" is coordinate (0,0) at layer 0 and "my top right corner" is coordinate (2,0) at layer 2.

To D2, "my bottom left corner" is coordinate (0,0) at layer 0 and "my top right corner" is coordinate (0,2) at layer 2.

To D3, "my bottom left corner" is coordinate (0,2) at layer 0 and "my top right corner" is coordinate (2,2) at layer 2.

EXAMPLE FRAME OF REFERENCE ANALYSIS:

Given board state

```
{
  "(0,0)": [],
  "(0,1)": [],
  "(0,2)": [],
  "(1,0)": [],
  "(1,1)": [],
  "(1,2)": [],
  "(2,0)": [],
  "(2,1)": [],
  "(2,2)": []
}
```

Given utterance

[D1: Could you please place a small orange block in my bottom left corner?]

Correct move

[PLACE:os:(0,0):0:CONFIRM:Placing small orange block at bottom-left of D1's side as requested]

Given utterance

[D2: Please remove the large orange block from my bottom left and middle cells.]

Correct move

[REMOVE:(0,0):0:(0,1):CONFIRM:Removing the large orange block from bottom-left+bottom-middle of D2's side as requested]

Given utterance

[D3: Let's begin by placing a large green block across the left and middle cells of my bottom layer.]

Correct move

[PLACE:gl:(0,2):0:(1,2):CONFIRM:Placing large green block across the left and middle cells of D3's bottom layer as requested]

Figure 12: Builder prompt used in the experiments, section I.

Builder Prompt (II): Board State, Oracle Candidates, and Execution Rules

Positions invisible to ALL directors: (1,1) and (2,1).

A large block that is visible to ANY of the directors CANNOT span EITHER (1,1) or (2,1).

— only inferred from what's missing in other views.

CURRENT BOARD STATE:

```
{json.dumps(current_state, indent=2)}
```

AVAILABLE BLOCKS:

```
{', '.join(available_blocks)}
```

BLOCK REFERENCE:

```
{block_reference}
```

COORDINATE REFERENCE:

```
{coordinate_reference}
```

CANDIDATE MOVES (verified physically valid for this turn):

```
{oracle_section}
```

DIRECTOR DISCUSSION:

```
{director_discussion}
```

DECISION RULE:

If 2+ directors agree on a block or position, do that first.

If all three disagree, pick the most specific instruction.

STACKING RULES:

- “layer” means stack depth, NOT grid row.
- ALWAYS calculate layer from CURRENT BOARD STATE, never trust director-specified layers.
- Before ANY place: count blocks at target position from CURRENT BOARD STATE.
 - You MUST place new blocks one layer above the number of blocks at that position (e.g., if position (0,1) has ['gs', 'o1'] → next block goes at layer 2; if position (0,1) has [''] → next block goes at layer 0).
- Before ANY remove: verify position is non-empty in CURRENT BOARD STATE.
 - If empty, do NOT attempt removal — tell directors and suggest placing instead.

FRAME OF REFERENCE RULE:

IMPORTANT: When choosing where to place a block, you MUST adopt the frame of reference of the director whose instruction you are following.

REMINDER: “The left” of D1’s view is coordinate (0,0) and “the right” is coordinate (2,0).

“The left” of D2’s view is coordinate (0,0) and “the right” is coordinate (0,2).

“The left” of D3’s view is coordinate (0,2) and “the right” is coordinate (2,2).

NEVER deviate from these frames of reference when executing instructions.

LARGE BLOCK RULE:

Large blocks span TWO adjacent cells — you MUST specify both endpoints.

To choose `span_to`:

- Identify the TWO director-relative cells explicitly referenced (e.g., “left+middle”, “middle+right”, “bottom left+bottom middle”).
- Convert those two cells into global coordinates using the DIRECTOR PERSPECTIVE GUIDE.
- Ensure BOTH cells lie on the correct wall for that director.
- Set `position` to one endpoint and `span_to` to the other endpoint.
- Before outputting, verify:
 - (a) `position` and `span_to` are orthogonal neighbors,
 - (b) both endpoint stacks have the SAME height (so placement/removal is on the same layer),
 - (c) neither endpoint is an invisible cell ((1,1) or (2,1)).

Figure 13: Builder prompt used in the experiments, section II.

Builder Prompt (III): Span Resolution, Failure Handling, and Output Format

NEVER place OR remove a large block if `span_to` is `None` — it will always fail.

Format:

`PLACE:block:position:layer:span_to:CONFIRM:reason`

Example:

`PLACE:g1:(0,0):0:(1,0):CONFIRM:Placing large green block across the left and middle cells of D1's bottom layer as requested`

If a director says "green large in the corner", you must figure out which two adjacent cells it spans from the CURRENT BOARD STATE.

NEVER place OR remove a large block if `span_to` is `None` — it will always fail.

If you try to remove a large block, you MUST check the board state to see where spans contain the same block.

EXAMPLE SPAN ANALYSIS:

Given board state

```
\texttt{\{
(0,0): [], (0,1): [], (0,2): []; \
(1,0): [], (1,1): [], (1,2): []; \
(2,0): [], (2,1): ["g1"], (2,2): ["g1"]
\}}
```

Given raw move

`[REMOVE:(2,2):0:CONFIRM:Removing the large green block from the bottom layer as requested by D3.]`

Correct move

`[REMOVE:(2,2):0:(2,1):CONFIRM:Removing the large green block from the bottom layer as requested by D3.]`

WHEN MOVES FAIL:

- Explain WHY: e.g., "I can't remove any block from the middle cell on the bottom layer. There is no block there. Suggest placing [block] instead."

- Never silently retry the same failed move

BEFORE PLACING:

Think step by step to make sure that you have interpreted the instructions, including block color and size, and the director's frame of reference, correctly. Do not place a block at the same place where you have previously removed a block of the same color. Count blocks at target position from CURRENT BOARD STATE.

EXAMPLE BLOCK COUNT AT TARGET POSITION:

Given board state:

```
\texttt{\{
(0,0): ["os"], (0,1): [], (0,2): []; \
(1,0): [], (1,1): [], (1,2): ["b1"]; \
(2,0): ["g1","b1"], (2,1): ["g1","b1"], (2,2): ["b1"]
\}}
```

Given raw move

`[PLACE:g1:(2,2):0:(2,1):CONFIRM:Placing large green block across the left and middle cells of D3's bottom layer as requested.]`

Correct move

`[PLACE:g1:(2,2):2:(2,1):CONFIRM:Placing large green block across the left and middle cells of D3's bottom layer as requested.]`

OUTPUT FORMAT — Choose ONE of these exact formats:

1. To place small block:

`PLACE:block_code:position:layer:CONFIRM:interpretation`

Example: `PLACE:bs:(0,0):0:CONFIRM:Placing blue small block at bottom-left of D1's side as requested`

2. To place large block:

`PLACE:block_code:position:layer:span_to:CONFIRM:interpretation`

Example: `PLACE:g1:(0,0):0:(1,0):CONFIRM:Placing large green block across left and middle cells of D1's bottom layer`

3. To remove small block:

`REMOVE:position:layer:CONFIRM:interpretation`

Example: `REMOVE:(1,2):0:CONFIRM:Removing the block from middle-right of D3's side as requested`

4. To remove large block:

`REMOVE:position:layer:span_to:CONFIRM:interpretation`

Example: `REMOVE:(2,2):0:(2,1):CONFIRM:Removing large green block from D3's bottom layer as requested`

NOTE: REMOVE never includes block code — do NOT write `REMOVE:b1:(0,0):...`

5. To clarify:

`CLARIFY:your specific question`

Example: `CLARIFY:Which blue block should I move - the one on top or bottom?`

Always include CONFIRM section to show what you understood from their instructions.

Figure 14: Builder prompt used in the experiments, section III.

Builder Prompt (IV): Tool Calling and Move Exploration

TOOL MODE — `simulate_move` available (`{max_simulations}` calls `max`):

WORKFLOW:

1. Simulate each director's instruction once directly and literally.
2. Pick the result with greatest value for "progress".
3. Submit that exact move as your FINAL answer. DO NOT INVENT NEW MOVE AFTER SIMULATING.
4. If a sim fails (`ok=False`) → fix ONLY the field the hint specifies, retry once.
5. NEVER submit a move that returned `ok=False`.
6. NEVER submit a remove move where simulate shows `structurePlacement=False` — even if it's the only `ok=True` simulation. In that case, CLARIFY instead.
7. NEVER clarify just because directors disagree — simulate and pick the best.
8. NEVER remove a block where simulate shows `structurePlacement=False` for that remove.

Figure 15: Builder’s More Exploration Tool Call Prompt. Note that although CRAFT provides this facility, this is not explored in our current benchmark, in favor of oracle moves in the Builder’s observation space in order to restrict the action space of the Builder for controlled experiments.

E LLM Judge Prompts and Experiments

Spatial Grounding Judge (SG) Evaluates the Director’s private `<think>` block in isolation, assessing whether S_{base_i} correctly identifies missing blocks, respects stacking constraints, and produces reasoning that corresponds to at least one oracle-correct move (Figure 16). Diagnoses F1 (limited search).

Spatial Grounding Judge (SG)
Given the director’s private target view, current board state, oracle correct moves, and internal reasoning, evaluate the spatial grounding quality of the director’s reasoning. SG1. Does the internal reasoning correctly identify at least one block missing from this director’s visible wall? SG2. Does the reasoning avoid describing blocks or positions already correctly placed on the board? SG3. Does the reasoning reference the correct layer for the missing block, accounting for what is already stacked? SG4. Does the reasoning identify at least one action matching or closely corresponding to an oracle correct move? SG5. Is the implied action executable given the current board state, respecting stacking order? SG6. Does the reasoning correctly interpret the size of the missing block (small vs. large) from the target view? SG7. Does the reasoning stay within this director’s visible wall cells rather than describing cells belonging to another director? Each question answered Yes / No / Unclear with a one-sentence justification. Response format: JSON with keys SG1–SG7, each containing answer and reason.

Figure 16: Spatial Grounding (SG) judge. Evaluated once per turn across 20 structures and 20 overall turns over the collective Director messages.

Mind Model Judge (MM) Evaluates the Director’s public `<message>` in the context of other Directors’ utterances and conversation history, assessing whether $L_{ToM_{joint}}$ produces a non-redundant, uniquely informative message the Builder can act on without clarification (Fig. 17). Diagnoses F2 (flawed pragmatics).

Pragmatic Sufficiency Judge (PS) Evaluates the collective Director output—all three public messages together—against the oracle candidate set, assessing whether the group jointly provided sufficient information for a rational Builder to identify a correct move without independent spatial reasoning (Fig. 18). Diagnoses the group-level communication failure defined in Sec. 3.1, irreducible to any individual Director’s SG or MM score.

Notably the three individual failure modes (F1–F3) from Nguyen [2024] (Section 3.1) are each **agent-local**: F1 (limited search) and F3 (inefficient inference) are evaluated per Director in isolation, while F2 (flawed pragmatics)—though its *target* is the joint listener $L_{ToM_{joint}}$ in CRAFT’s multi-sender setting—is still *measured* per Director through the MM judge, which evaluates each Director’s message individually for epistemic calibration. The fourth failure mode introduced in Sec. 3.1 is categorically different: it is both measured and defined at the group level, occurring when all three Directors individually produce non-redundant, calibrated messages (F2 absent) yet their collective output still fails to identify an oracle move for the Builder. No individual Director’s SG or MM score can detect this failure—it is only measurable through the PS judge, which evaluates the full collective Director output against the oracle candidate set.

Mind Model Judge (MM)
Given the director’s internal reasoning, public message, other directors’ messages, and recent conversation history, evaluate the Theory-of-Mind quality of the director’s public communication. MM1. Does the public message add information not already communicated by other directors in this or the immediately preceding turn? MM2. Does the message avoid repeating an instruction already given and acted upon in a previous turn? MM3. Does the message reflect awareness of what the builder already knows from the conversation history? MM4. Does the message accurately translate the key finding from internal reasoning into natural language without losing critical spatial information? MM5. Does the message focus on information uniquely visible from this director’s wall rather than information another director could equally provide? MM6. Is the message specific enough for the builder to execute without further clarification, naming a block type, location, and action? MM7. If another director gave a conflicting instruction this turn, does the message acknowledge or attempt to resolve the conflict? MM8. Does the message show awareness of the boundary between what this director uniquely sees and what other directors can also see? Each question answered Yes / No / Unclear with a one-sentence justification. Response format: JSON with keys MM1–MM8, each containing answer and reason.

Figure 17: Mind Model (MM) Judge. Evaluated once per turn across 20 structures and 20 overall turns over the collective Director messages. Note that oracle correct moves are not provided to the MM judge during evaluation.

Pragmatic Sufficiency Judge (PS)
Given the current board state, oracle correct moves, director messages, and builder confirmation, evaluate whether the collective director messages were pragmatically sufficient to guide the builder toward a correct move. PS1. Do the director messages collectively identify at least one specific location on the board that needs a block placed or removed? PS2. Do the director messages collectively specify the correct block type — both color <i>and</i> size (small vs. large) — for at least one oracle correct move? PS3. Would a rational builder reading only these messages have sufficient information to select at least one oracle correct move <i>without</i> independent spatial reasoning about the target structure? PS4. Do the messages use precise spatial anchors that uniquely identify the target location, rather than vague relative language that could map to multiple grid positions? PS5. Does the builder confirmation indicate it correctly understood the directors’ collective intent, regardless of whether execution succeeded? PS6. If the builder did not execute the correct move, was the failure primarily attributable to director underspecification rather than builder execution mechanics (wrong layer, missing span, stacking violation)? <i>N/A if oracle move was executed.</i> Each question answered Yes / No / Unclear (or N/A for PS6) with a one-sentence justification. Response format: JSON with keys PS1–PS6, each containing answer and reason.

Figure 18: Pragmatic Sufficiency (PS) judge. Evaluated once per turn over the collective Director messages. Six binary questions targeting location specificity, block type precision, rational sufficiency, spatial anchoring, Builder understanding, and failure attribution.

Judge	Q	Exact	W. κ	Rank ρ	p
SG	SG1	0.665	0.265	0.821	0.000
	SG2	0.547	0.146	0.697	0.004
	SG3	0.473	0.260	0.516	0.049
	SG4	0.510	0.139	0.427	0.113
	SG5	0.565	0.314	0.649	0.009
	SG6	0.545	0.271	0.825	0.000
	SG7	0.902	0.570	0.772	0.001
MM	MM1	0.697	0.294	0.563	0.029
	MM2	0.565	0.181	0.226	0.418
	MM3	0.360	0.039	0.496	0.060
	MM4	0.835	0.456	0.231	0.407
	MM5	0.723	0.456	0.753	0.001
	MM6	0.827	0.568	0.604	0.017
	MM7	0.810	0.252	0.682	0.005
PS	PS1	0.864	0.413	0.380	0.163
	PS2	0.736	0.365	0.402	0.138
	PS3	0.780	0.413	0.555	0.032
	PS4	0.888	0.373	0.467	0.079
	PS5	0.736	0.553	0.052	0.855
	PS6	0.956	0.898	0.619	0.014
Overall	SG	0.601	0.281	0.672	—
	MM	0.678	0.403	0.528	—
	PS	0.827	0.503	0.412	—

Table 6: **GPT-4o-mini vs GPT-4o agreement** on a stratified subsample selected as 2 turns per model $\times$ structure for SG and MM (15 models $\times$ 20 structures $\times$ 2 turns = 600 director-turns each, yielding 4,200 and 4,800 question-answer pairs) and 1 turn per model $\times$ structure for PS ($n = 295$ turns, 1,770 pairs). Exact agreement and linearly weighted κ measure absolute agreement; Rank ρ reports Spearman correlation of per-model mean scores between judges ($n = 15$ models). **Bold** rows indicate questions with significant rank preservation ($p < 0.05$). GPT-4o applies stricter correctness thresholds throughout, producing lower Yes rates; the preserved model rankings confirm that relative comparisons reported in the paper are robust to this calibration difference.

E.1 LLM Grader Validation

Due to the scale of CRAFT evaluation—thousands of turn logs across 15 Director models, 20 structures, and multiple runs—we first validated GPT-4o-mini against GPT-4o on a stratified subsample (600 director-turns for SG/MM and 295 for PS). Table 6 reports exact agreement, weighted κ , and Spearman rank correlation between the two judges. Agreement is strongest on the questions central to our analysis: MM5 (unique perspective utilization) achieves $\rho = 0.753$ ($p = 0.001$), SG7 (wall adherence) $\rho = 0.772$ ($p = 0.001$), and PS3 (rational sufficiency) $\rho = 0.555$ ($p = 0.032$), with PS6 (failure attribution) reaching near-perfect agreement ($\kappa = 0.898$). **Overall, model rankings are consistently preserved across key diagnostic dimensions, indicating that GPT-4o-mini (vs GPT4o) reliably captures the relative differences between models.** Based on this validation and for computational cost-efficiency, we use GPT-4o-mini for full-scale evaluation reported in Figure 5 and Figure 19, enabling tractable large-scale analysis while maintaining alignment with a stronger judge on crucial metrics in our experiments.

Additionally, Table 7 shows the mean LLM grader scores using our primary grader model (GPT4o-mini) for our main results—averaged across 3 independent runs. The low SEM values show that graders evaluate consistently. Fig. 19 shows per-model LLM grader scores across all individual judge questions, showing where individual model failures may occur. Table 8 shows Pearson and Spearman correlations of Director communication features with task progress at turn 20, showing that a specific subset of features, such as the *remove gap* have strong *negative* correlation with task progress, even if the features themselves imply better individual communication, such as introducing novel information or leveraging the Director’s unique individual perspective view.

Model	SG $\pm$ SEM	MM $\pm$ SEM	PS $\pm$ SEM	Progress
Frontier Models				
Gemini-3-Flash	0.669 ± 0.010	0.237 ± 0.009	0.356 ± 0.015	0.675
GPT-4o	0.792 ± 0.006	0.706 ± 0.004	0.492 ± 0.010	0.588
GPT-4o-Mini	0.741 ± 0.005	0.632 ± 0.005	0.457 ± 0.006	0.333
GPT-4.1-Mini	0.937 ± 0.003	0.787 ± 0.003	0.481 ± 0.005	0.312
Claude-Sonnet-4.6	0.910 ± 0.006	0.775 ± 0.006	0.444 ± 0.006	0.285
Gemini-2.5-Flash	0.932 ± 0.003	0.757 ± 0.006	0.479 ± 0.011	0.257
Gemini-3.1-Flash-Lite-Preview	0.819 ± 0.006	0.597 ± 0.007	0.481 ± 0.008	0.257
Open-Weight Models				
Mistral-7B	0.716 ± 0.007	0.555 ± 0.005	0.509 ± 0.009	0.631
Qwen-7B	0.753 ± 0.006	0.530 ± 0.005	0.514 ± 0.012	0.612
Llama-8B	0.744 ± 0.006	0.491 ± 0.006	0.500 ± 0.011	0.586
Gemma-9B	0.673 ± 0.007	0.507 ± 0.005	0.481 ± 0.010	0.578
Qwen-72B	0.745 ± 0.007	0.503 ± 0.008	0.378 ± 0.010	0.557
Qwen-14B	0.748 ± 0.005	0.523 ± 0.004	0.511 ± 0.008	0.476
DeepSeek-Lite	0.142 ± 0.005	0.334 ± 0.004	0.617 ± 0.011	0.419
Qwen-32B	0.742 ± 0.006	0.571 ± 0.005	0.476 ± 0.006	0.339
Group Averages				
Open-Weight	0.658 ± 0.069	0.502 ± 0.024	0.499 ± 0.022	0.525
Frontier	0.829 ± 0.036	0.642 ± 0.067	0.452 ± 0.019	0.387

Table 7: LLM grader scores (mean $\pm$ SEM) for spatial grounding (SG), mind modeling (MM), and pragmatic sufficiency (PS), alongside task progress. Frontier models achieve higher SG and MM scores but lower progress than open-weight models. All scores are averaged over **three independent grader runs**.

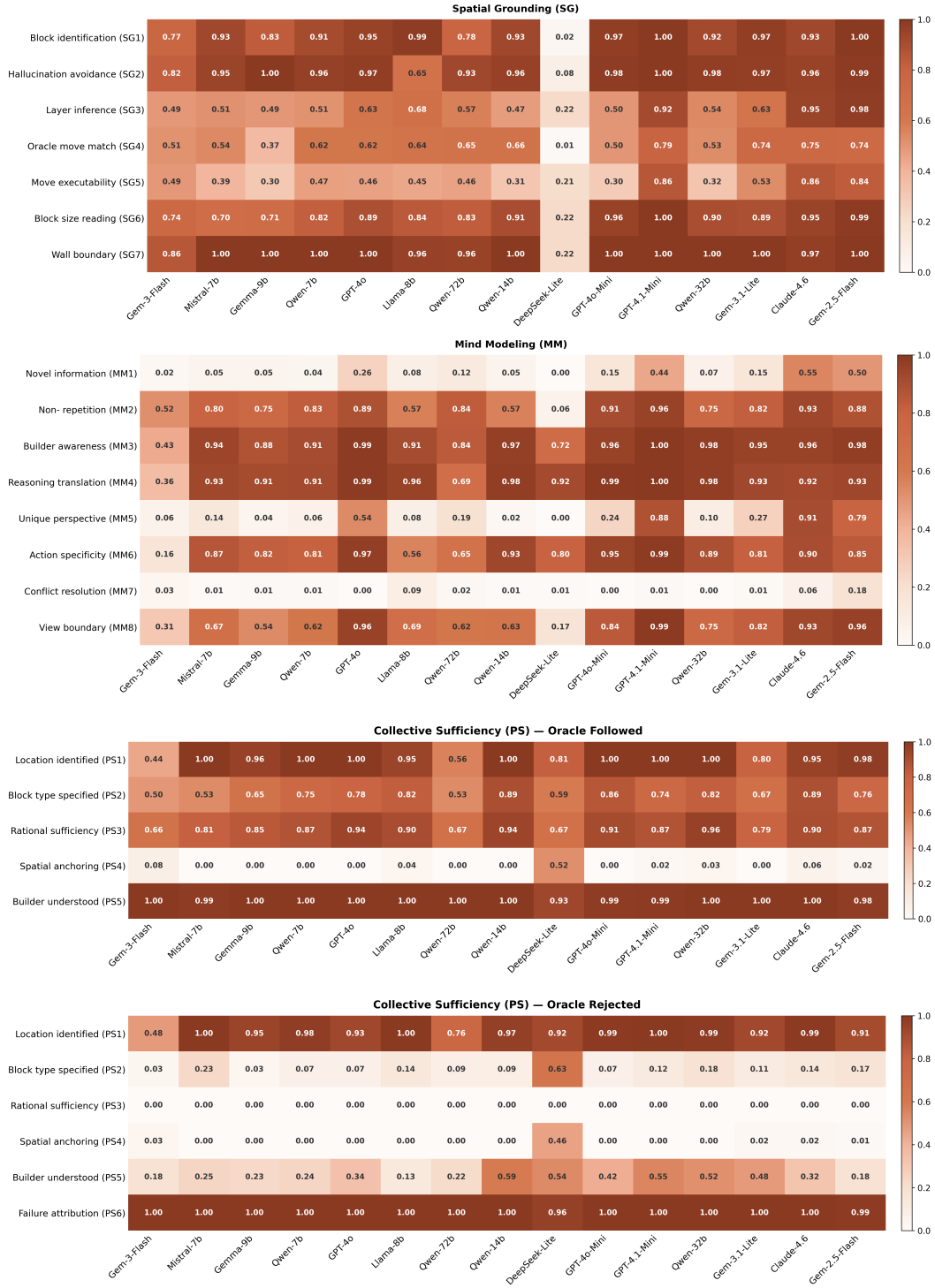


Figure 19: Per-model LLM grader scores across all judge questions for spatial grounding (top), mind modeling (second), and pragmatic sufficiency split by condition—oracle followed (third) and oracle rejected (bottom). Models are sorted by task progress descending along the x-axis. PS scores for oracle-followed turns exclude PS6 (Failure attribution), which is not applicable when the builder executes a correct move. All scores are averaged across independent grader runs ($n=3$).

Feature	Pearson r	p	Spearman ρ	p
<i>Behavioral</i>				
Remove gap	−0.780***	0.001	−0.785***	0.001
<i>Message calibration (MM)</i>				
Novel information (MM1)	−0.619*	0.014	−0.652**	0.008
Overall MM score	−0.603*	0.017	−0.613*	0.015
Unique perspective (MM5)	−0.575*	0.025	−0.536*	0.039
Action specificity (MM6)	−0.514	0.050	−0.311	0.259
Conflict resolution (MM7)	−0.284	0.304	−0.199	0.478
<i>Spatial reasoning (SG)</i>				
Layer inference (SG3)	−0.483	0.068	−0.472	0.076
Move executability (SG5)	−0.443	0.098	−0.302	0.274
Overall SG score	−0.273	0.326	−0.554*	0.032
<i>Collective (PS)</i>				
Rational sufficiency (PS3)	+0.666**	0.007	+0.633*	0.011
<i>Surface</i>				
Message length (words)	+0.046	0.870	−0.295	0.286
<i>Mediation: unique perspective → remove gap → progress</i>				
Unique perspective alone	$R^2 = 0.330$			
Remove gap alone	$R^2 = 0.609$			
Unique perspective + remove gap	$R^2 = 0.633$	$(\Delta R^2 = 0.024)$		
Partial r (unique perspective remove gap)	$r = -0.247$	$p = 0.374$ (n.s.)		

Table 8: Pearson and Spearman correlations of Director communication features with task progress at turn 20 ($n = 15$ models). Features are ordered by absolute Pearson r . Judge question codes in parentheses refer to the evaluation framework in Sec. E. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

E.2 Reconciling Task Performance with LLM Judge results via “Remove-gap” analysis

Among the features in Table 8, MM5 and PS3 stand out as the strongest individual-level and group-level predictors of task progress respectively—and as the only features that directly implicate the correction spiral mechanism through their relationship with remove gap—motivating a mediation analysis for each. Specifically, to test whether remove gap mediates the relationship between unique perspective utilization (MM5) and task progress, we regressed task progress on MM5 alone ($R^2 = 0.330$), remove gap alone ($R^2 = 0.609$), and both jointly ($R^2 = 0.633$). Adding MM5 to remove gap increases explained variance by only 2.4 percentage points, and the partial correlation between MM5 and progress controlling for remove gap is non-significant ($r = -0.247$, $p = 0.374$). Remove gap fully mediates the MM5–progress relationship: Directors who leverage their unique perspective produce more correction-oriented instructions, driving over-removal that consumes the turn budget without advancing task progress. The direct effect of communication quality on task outcomes is entirely absorbed by the behavioral signature it produces overall.

Collective Pragmatic Sufficiency as an Independent Path. Collective pragmatic sufficiency (PS3) reveals a second, independent upstream path to the same mediator. PS3 directly predicts remove gap ($r = -0.565$, $p = 0.028$) and task progress ($r = +0.666$, $p = 0.007$; $\rho = +0.633$, $p = 0.011$), and remains a significant predictor of progress after controlling for MM score (partial $r = 0.628$, $p = 0.012$). MM5 and PS3 are not significantly correlated ($r = -0.363$, $p = 0.183$) and explain complementary variance in remove gap (MM5 alone $R^2 = 0.329$, PS3 alone $R^2 = 0.319$, jointly $R^2 = 0.475$), confirming that they operate through distinct mechanisms.

In contrast to MM5 pathway which is *proactive*, the PS3 path is *reactive*: when collective Director output fails to identify an oracle move, the Builder makes incorrect placements that accumulate board errors, subsequently triggering removal instructions. Both paths converge on remove gap as the proximal behavioral mechanism, consistent with the full mediation established above.

Correction Spirals: A Frontier Model-specific Failure Mode Importantly, splitting by model group reveals that this mediation is *asymmetric*. For **frontier** models ($n = 7$), remove gap alone explains $R^2 = 0.941$ of progress variance ($r = -0.970$, $p = 0.006$), with full mediation of MM5’s effect confirmed (partial $r = 0.098$, $p = 0.876$ after controlling for remove gap). For **open-weight** models ($n = 8$), remove gap explains only $R^2 = 0.286$ of progress variance ($r = -0.535$, $p = 0.172$), and MM5 shows no significant relationship with remove gap ($r = 0.205$, $p = 0.627$). Notably, MM5 correlates *positively* with progress in open-weight models ($r = +0.312$)—the opposite direction from frontier models—indicating that unique perspective utilization does not activate the correction spiral behavior when epistemic calibration is weaker overall. The correction spiral is therefore a **frontier-specific failure mode**: frontier models’ superior ability to leverage unique wall perspective produces correction-oriented instructions that drive over-removal, in a way that open-weight models’ weaker epistemic calibration does not.

Ruling Out Context Availability as a Confound in Epistemic Uniqueness (MM5). A potential confound in the frontier–open-weight performance asymmetry is that smaller models may receive insufficient context to reason about epistemic uniqueness. We rule this out directly: all Directors receive *identical*¹³ inputs at each turn—the same partial target view, current board state, and full conversation history. The information needed to determine what is uniquely visible from one’s wall—i.e., what co-Directors have already communicated and what remains exclusively within one’s view—is equally available to all models. Thus, the gap in MM5 scores reflects differences in *reasoning over* context rather than *access to* it. This implies that the correction spiral is not an artifact of context limitations or information bottlenecks, but a *capability threshold effect*: sufficient epistemic calibration is required to translate available context into uniquely grounded correction instructions. Open-weight models fall below this threshold not because they see less, but because they cannot reliably act on shared context in an epistemically calibrated way. CRAFT’s physically constrained environment where over-correction directly impacts task progress, enables this threshold effect to surface clearly, rather than being masked in more permissive evaluation settings.

¹³We do not mean identical in that every game contains the same inputs exactly, but that the information available to directors is consistent across models during game runs.

F Additional Experimental Results and Examples

F.1 Connection Between Homogeneity of Director Groups and Equal Weighting in joint ToM listener (via λ_i in Eq. 3).

As noted in Section A, we perform an empirical analysis of per-Director influence using Builder attribution or the “credit-rate” on oracle-correct turns, i.e., instances where the Builder executes a move matching the oracle, which we treat as reliable signals of task-aligned behavior (see Appendix B). We compute normalized attribution weights $\hat{\lambda}_i$ based on which Director’s instruction is followed during successful execution. **As shown in Table 9, $\hat{\lambda}_i$ are approximately uniform across models, indicating that the Builder integrates Director signals symmetrically in this setting.** Importantly, these estimates reflect *observed behavioral influence* rather than counterfactual causal credit. Nevertheless, their near-uniformity provides empirical support for the equal-weighting behavior implied by symmetry, while deviations in heterogeneous settings would indicate meaningful differences in agent influence and motivate learning λ_i .

Specifically, for each oracle-followed turn, we parse the Builder’s confirmation text to identify *which* Directors received attribution credit. We define the credit rate for Director D_i as the fraction of oracle-followed turns in which the Builder’s confirmation explicitly references D_i , normalized by D_i ’s participation rate across all turns (mean ≈ 0.70 per Director, approximately equal across all 15 models—confirming that participation frequency does not confound the analysis).

For a robust analysis, we additionally apply a degeneracy filter to remove turns where Director messages are empty or truncated, raw board state JSON dumps, or near-duplicate copies of other Directors’ messages. Note that progress metrics and LLM grader runs/scores are computed over *all* logged turns to capture the full distribution of Director behavior including failures. In contrast, the λ_i credit attribution is restricted to oracle-correct turns (where the builder followed the oracle-suggested move exactly) and clean Director messages because only on these turns does Builder confirmation text reflect a meaningful attribution of instruction credit to a specific Director. Specifically, empirical λ_i values are then computed by normalizing the clean credit rates to sum to one per model. As seen in Table 9, empirical λ_i values are close to equal weighting across all 15 models, with mean values of $\lambda_{D1} = 0.332$, $\lambda_{D2} = 0.352$, and $\lambda_{D3} = 0.316$ overall. Deviations from equal weighting ($\lambda = 0.333$) are small and consistent: $\Delta\lambda_{D2} = +0.019$, $\Delta\lambda_{D1} = -0.001$, $\Delta\lambda_{D3} = -0.017$, with the same pattern holding for both open-weight and frontier groups. This validates the equal λ_i assumption as a reasonable approximation for the homogeneous Director configuration used in this work.

Structural Information Geometry. The small but consistent deviation $\lambda_{D2} > \lambda_{D1} > \lambda_{D3}$ is structurally interpretable at the aggregate level. Averaged across models, Director D_2 ’s far-wall view covers 51.7% of oracle-selected positions—the highest among the three—including interior cells (1, 1) and (2, 1) that are uniquely visible to D_2 . In comparison, D_1 and D_3 cover 44.1% and 36.5%, respectively. Despite this advantage, the Builder does not fully exploit D_2 ’s structural coverage: its average weight ($\lambda_{D2} \approx 0.352$) remains substantially below its coverage (0.517), while D_3 ’s weight ($\lambda_{D3} \approx 0.316$) is closest to its coverage (0.365, gap -0.049). This indicates that the Builder distributes influence more evenly than the underlying information geometry would suggest, reflecting a conservative aggregation strategy that limits dominance by any single Director despite asymmetric visibility. This is consistent with prior work [Riedl, 2025] that suggests that group “synergy” can reliably emerge in complex multi-agent systems beyond structural constraints encoded in the task setup.

Remove-Gap Evolution Across Turns Figs. 20 and 21 show evolution of Builder remove actions across turns against oracle-prescribed remove actions across all evaluations for all models. This comparison highlights the *over-removal* that proprietary frontier models typically engage in when compared to open-weight models. Even larger open-weight models engage in over-removal, leading to *correction spirals* (Fig. 22, Table 11).

Failed Moves During our evaluation runs on 20 target structures, we found 47 turns with no recorded move attempt (30 for base models, 17 for proprietary models). Inspection of the raw logs revealed these are attributable to malformed or empty Director responses that left the Builder with insufficient information to construct a valid action, rather than any systematic model failure.

Model	λ_{D1}	λ_{D2}	λ_{D3}	λ -std	Filtered	Total	Oracle	Orc%	Deg%
Frontier Models									
Claude-Sonnet-4.6	0.336	0.363	0.302	0.0250	35	400	176	44.0%	11.8%
Gemini-2.5-Flash	0.340	0.351	0.308	0.0182	48	400	164	41.0%	15.7%
Gemini-3-Flash	0.344	0.360	0.296	0.0270	84	385	269	69.9%	21.3%
Gemini-3.1-Flash-Lite-Preview	0.323	0.347	0.331	0.0100	28	400	158	39.5%	10.1%
GPT-4.1-Mini	0.334	0.320	0.346	0.0106	26	400	171	42.8%	8.9%
GPT-4o-Mini	0.342	0.341	0.317	0.0118	39	400	192	48.0%	11.6%
GPT-4o	0.317	0.376	0.307	0.0306	47	397	262	66.0%	10.5%
Open-weight Models									
DeepSeek-Lite	0.334	0.354	0.311	0.0174	150	400	248	62.0%	33.6%
Gemma-9b	0.325	0.342	0.333	0.0067	52	400	245	61.3%	12.9%
Llama-8b	0.316	0.368	0.317	0.0244	89	396	260	65.7%	20.1%
Mistral-7b	0.315	0.373	0.312	0.0281	62	388	258	66.5%	13.2%
Qwen-14b	0.340	0.360	0.300	0.0253	51	391	218	55.8%	12.1%
Qwen-32b	0.327	0.336	0.338	0.0049	30	400	167	41.8%	10.5%
Qwen-72b	0.325	0.361	0.314	0.0200	61	399	244	61.2%	15.2%
Qwen-7b	0.318	0.352	0.331	0.0141	66	390	268	68.7%	13.0%

Table 9: Empirical λ_i weights and associated diagnostics. λ_{D1} , λ_{D2} , λ_{D3} denote the empirical credit share attributed to each Director by the Builder on oracle-correct turns, normalized to sum to one. **Oracle** is the number of turns where the Builder executed a verified oracle-correct move. **Orc%** is the fraction of total turns corresponding to oracle-correct moves. **Deg%** denotes the percentage of degenerate credits removed during filtering. Total represents the total number of actual turns evaluated (termination condition) across 20 test structures.

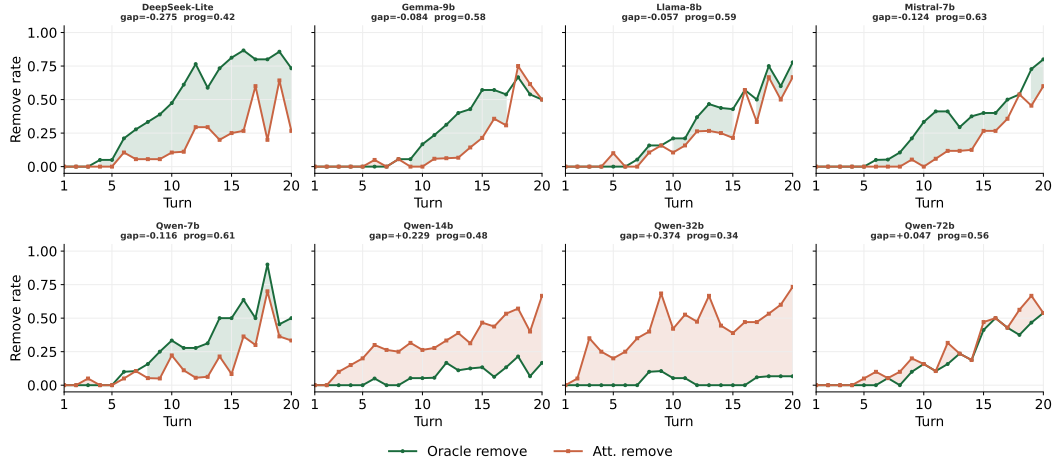


Figure 20: Oracle-prescribed vs. attempted remove rate per turn for all base open-weight models.

Qwen-72b accounts for the majority of base model cases (23 of 30). These turns are excluded for the involved models in the FAIL or failed move rates in Table 1. Additionally, the Builder agent’s clarification rate is negligible across all models—at most one CLARIFY response per model across 385–400 turns (rate ≤ 0.003)—confirming that Builders consistently attempt moves rather than request additional information, even when Director instructions could be ambiguous.

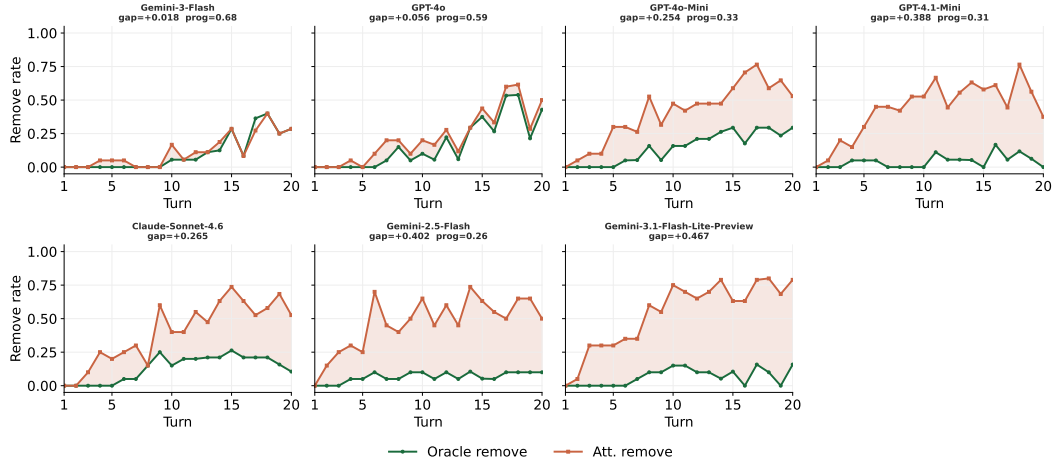


Figure 21: Oracle-prescribed vs. attempted remove rate per turn for all frontier and proprietary models.

Model	Orc_Remove↓	Att_Remove	Gap
<i>Open-weight models</i>			
DeepSeek-Lite	0.433	0.148	<u>−0.275</u>
Mistral-7b	0.238	0.124	−0.124
Qwen-7b	0.238	0.205	−0.116
Gemma-9b	0.211	0.122	−0.084
Llama-8b	0.232	0.277	−0.057
Qwen-72b	0.162	0.245	+0.047
Qwen-14b	0.062	0.355	+0.229
Qwen-32b	0.027	0.448	+0.374
<i>Frontier models</i>			
Gemini-3-Flash	0.073	0.196	+0.018
GPT-4o	0.146	0.280	+0.056
Claude-Son.-4.6	0.130	0.395	+0.265
GPT-4o-Mini	0.139	0.432	+0.254
GPT-4.1-Mini	0.040	0.462	+0.388
Gemini-2.5-Flash	0.065	0.468	<u>+0.402</u>
Gemini-3.1-Flash-Lite	0.066	0.540	<u>+0.467</u>
Overall	0.148	0.314	+0.135
<i>Turn-level Spearman ρ vs oracle adherence</i>			
Gap	$\rho = -0.543, p < 0.001$		
Att_Remove	$\rho = -0.569, p < 0.001$		

Table 10: Remove action rates per model. Oracle remove rate reflects what the CRAFT game engine considers necessary given the current board state; attempted remove rate reflects what Directors instructed the Builder to do. Negative gap indicates under-removal (Builder removes less than oracle prescribes); positive gap indicates over-removal. Bold = best within group. Underline = largest absolute gap. Turn-level Spearman correlations use full oracle adherence as outcome ($n = 5,382$).

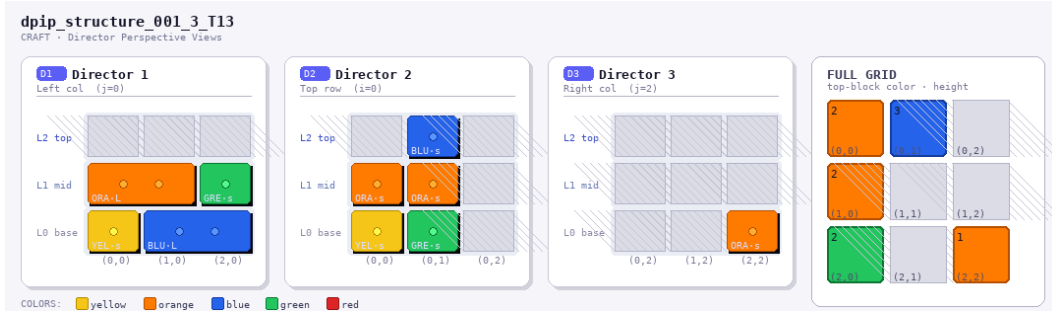


Figure 22: **Three turns, zero progress: a CRAFT correction spiral** (Qwen-32b, structure_001, T10–T14) (see Table 11 for the full turn-by-turn trace). D1 (left wall) sees a large orange domino at layer 1 spanning $(0,0) - (1,0)$ sitting directly on top of a small yellow at $(0,0)$ layer 0 and a large blue domino spanning $(1,0) - (2,0)$ at layer 0. The orange domino is therefore the *top* block at both positions and is directly accessible for removal — yet D1 instructs “remove the large orange block from the *bottom* layer,” targeting layer 0 which is occupied by the blue domino. D2 (far wall) independently sees orange small blocks at layer 1 on $(0,0)$ and $(0,1)$ but similarly instructs removal from “bottom left corner,” again specifying layer 0. D3 (right wall) sees only a single orange block at $(2,2)$ layer 0 and has no visibility into either problem position, remaining entirely unaware of the “deadlock”. The full grid minimap shows five occupied positions with maximum height 3 at $(0,1)$; the orange blocks at $(0,0)$ and $(1,0)$ each sit at height 2 as the topmost block, confirming they are accessible. Despite this, neither D1 nor D2 corrects the layer specification across turns 10, 13, and 14—the Builder’s oracle recommends forward placement moves throughout (`place gs @ (0,0) L2` , `place rl @ (1,0) L2`) but the **Builder follows the Director consensus** and attempts the *same* failed remove three times, consuming turns without any board state change.

Turn	Oracle (top move)	Director messages & outcome
T10	place gs @(0,0) L2	D1: “remove the large orange block from the middle-left of my bottom layer ” · D2: “remove orange from bottom left corner” · Builder follows D1 → Cannot remove layer 0 at (1,0)
T11	place gs @(0,0) L2	D2: same remove instruction · D3: “confirm small orange in bottom right” · Builder follows D3 → remove os @(0,2) L0 ✓ progress=0.355
T12	place gs @(0,0) L2	D1: same remove instruction · D3: “add small orange to bottom right” · Builder follows D3 → place os @(2,2) L0 ✓ progress=0.386
T13	place gs @(0,0) L2	D1: “get rid of large orange from bottom layer middle-left” · D2: “remove orange from bottom left corner ” · D3: unaware of deadlock, confirms own wall · Builder follows D1+D2 consensus → Cannot remove layer 0 at (1,0)
T14	place gs @(0,0) L2	D1: identical instruction as T13 · D3: “focus on removing the large orange from D1’s bottom layer” (endorses wrong instruction) · Builder follows D1+D3 consensus → Cannot remove layer 0 at (1,0)

Board state at (1,0) is [b1, o1] throughout T10–T14: b1 at layer 0, o1 at layer 1. The orange block is accessible at layer 1 but directors consistently specify layer 0. Oracle recommends placement moves throughout; no director ever issues the correct prerequisite remove o1 @(1,0) L1.

Table 11: Turn-by-turn trace of the Qwen-32b correction spiral on structure_001. Despite the oracle recommending forward placement moves at every turn, D1 and D2 repeatedly issue remove instructions targeting the wrong layer, while D3 remains unaware of the deadlock. The Builder escapes the spiral on T11 and T12 by following D3 instead, but returns to the failed pattern when D1 and D2 reach consensus on T13–14.

G Experimental Configuration

Hardware Setup and Decoding Settings All open-weight Director models were served locally on a machine with two NVIDIA RTX PRO 6000 Blackwell GPUs (96 GB VRAM each), using bfloat16 precision with 4-bit quantization for Qwen-32B and Qwen-72B. A single 20-turn game on one structure took approximately 5 minutes for smaller open-weight models (e.g., Qwen-7B, Llama-8B) and up to 15 minutes for the 72B model. Frontier API models were called sequentially with per-model rate-limiting; a single GPT-4o or GPT-4o-Mini run (20 turns) over one structure took approximately 12 minutes owing to API latency across 20 turns. All runs used a fixed random seed per structure-run index pair for reproducibility, with each model evaluated with LLM graders across 3 independent runs and 20 structures. Open-weight models use greedy decoding (temperature=0) for reproducibility, since local inference is fully deterministic; frontier API models use temperature=0.7 as API providers do not guarantee deterministic outputs even at temperature=0 [OpenAI, 2023a].¹⁴.

Termination condition In CRAFT, games are terminated early upon full structure completion to save additional compute needed for redundant turns; across all 15 models, 16 of 300 total game runs (5.3%) reached full completion before turn 20, with Gemini-3-Flash and Mistral-7B achieving the most early completions (4 each), consistent with their leading task progress scores. Where the conversation history accumulates > 50 total messages during a game, we truncate it to 40 previous messages for the remaining turns to reduce context-exposure [Hong et al.] and cost.

¹⁴<https://152334h.github.io/blog/non-determinism-in-gpt-4/>